

Unlabeled circle graphs (OEIS A156808 and A156809)

The leading equivalent with error bounds and inverses,
and the first corrections

Two research reports of one external research session
(bundle Reports 150 and 152), merged into one ProveIt report

Parts I and II: 3 October 2026 merged: 5 October 2026

Abstract

Let c_n and g_n count connected and all unlabeled circle graphs on n vertices (OEIS A156808 and A156809), and put $h_n = (2n - 1)!!/(4n)$. This report prints two manuscripts as Parts I and II, the second a sequel to the first. Part I proves $c_n \sim g_n \sim e^{-3/2}h_n$, matching an upper bound from reciprocal representation fibres with an injective construction from asymmetric split-prime cores carrying leaf, true-twin and false-twin decorations. It also proves the connected fraction $1 - 1/(2n) + O(n^{-2})$, relative errors $O(1/n)$ for both counts through a self-contained Poisson approximation of local matching patterns, independent limiting Poisson(1/2) decoration counts, a Poisson(3/2) core deficit, the asymmetry probability e^{-1} , a labeled equivalent, and a Lambert- W inverse with ceiling-safe brackets; and it derives A156808(13) = 21,593,488,017 from the OEIS value of A156809(13). Its prime probability $e^{-3} + o(1)$ follows from lemmas of Bassino, Bouvel, Féray, Gerin and Pierrot, who state the bound e^{-4} ; Part I says that this is not a new theorem. Part II proves $c_n = e^{-3/2}h_n(1 - \frac{15}{4n} + O(n^{-2}))$, $g_n = e^{-3/2}h_n(1 - \frac{13}{4n} + O(n^{-2}))$ and $c_n/g_n = 1 - \frac{1}{2n} - \frac{1}{n^2} + O(n^{-3})$, from a uniform local expansion that gives the prime probability $e^{-3}(1 - 5/n + O(n^{-2}))$ and a canonical split-tree transfer to graph classes. It adds first corrections for automorphism-weighted and labeled counts and for rare automorphism groups, a refined inverse, and a transfer theorem for every fixed order that is conditional on the corresponding expansion of the prime counts; its second-order coefficients depend on an unknown prime constant β . The value of β , an unconditional expansion beyond the first correction and effective constants remain open. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 47fc7a069 (batch 106). They study one pair of sequences, the numbers c_n (OEIS A156808) and g_n (OEIS A156809) of connected and of all unlabeled circle graphs of order n , in one model: the indexed perfect matchings of $2n$ cyclically ordered endpoints, $m_n = (2n - 1)!!$ of them, acted on by the dihedral group of order $4n$, with the normalization $h_n = m_n/(4n)$.

Part	Bundle report	Delivered title	Labels
I	150 (base)	Leading asymptotics of unlabeled circle graphs	cgr:ld:
II	152	First corrections for unlabeled circle graphs	cgr:fc:

Arrangement. Report 150 is the base of the merge (its `Report150.tex` was staged as this `article.tex`) and is printed first, as Part I: it introduces the setting, proves the leading equivalent and its relative $O(1/n)$ error, and alone proves the typical-core laws, the explicit local total-variation bound and the derived value c_{13} . Report 152 is a sequel, not a new edition: it says that “the earlier Report 150 established the leading equivalent and relative $O(1/n)$ bounds; this report identifies the individual first coefficients” (Section 15.2). It is printed second, as Part II. Its archive carries no copy of Report 150. Dependency order is also the order of writing.

Numbering. Sections are numbered continuously; statements and equations are numbered within sections, as in both manuscripts. Part I keeps Report 150’s Sections 1–12, statement and equation numbers; its Appendix A is Section 13. In Part II a section number is Report 152’s plus 13 (Sections 14–24), and so are the first components of its statement and equation numbers: its Theorems 1.1 and 1.2 are Theorems 14.1 and 14.2, its Theorem 3.1 is Theorem 16.1, its Lemma 5.1 is Lemma 18.1, its Theorem 6.1 is Theorem 19.1, its Theorem 10.1 is Theorem 23.1, and its equation $(k.j)$ is equation $((k + 13).j)$ here. Remark 3.3, the last numbered statement of its section, and Section 25, which closes the report, were added in the write; no delivered number moved.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date (5 October 2026). They give cross-references between the Parts, mark statements that Part II supersedes or answers, credit prior work, and record what the write checked. Superseded statements and answered questions are printed as delivered, with a dated note; no delivered statement was changed and no delivered symbol was renamed. Part I’s open questions are its Section 12; Part II states its limits in its last Section 24; Section 25 gathers both under Vladimir’s standing rule of 4 October 2026, with their sources, sketches and what is missing.

What is proved, and where

Here $m_n = (2n - 1)!!$, $h_n = m_n/(4n)$, p_n^a counts asymmetric split-prime circle graphs of order n , and $u = L/W_0(2L/e)$ with L the logarithm of the threshold.

Statement	Where
$c_n \sim g_n \sim e^{-3/2} h_n$; $g_n - c_n = g_{n-1} + O(m_n/n^3)$; $c_n/g_n = 1 - \frac{1}{2n} + o(n^{-1})$ Prime probability $e^{-3} + o(1)$ (the source states $> e^{-4}$) $S_n \leq 4n(2en)^{n/2} e^{\sqrt{n}}$ for the symmetric matchings	Part I, Theorem 1.1 (Sections 3–7) Part I, (2.6), from [1, Lemmas 5.7–5.10] Part I, Lemma 3.1; restated in Part II, (15.2); method of [1, Proposition 5.14] ([write] Remark 3.3) Part I, Theorem 8.1, Proposition 8.2, (8.6), (8.8)
Typical recovered core: decoration counts $\rightarrow \text{Poisson}(1/2)^{\otimes 3}$, deficit $\rightarrow \text{Poisson}(3/2)$; $\text{Aut}(G) \cong (C_2)^{t+f}$; asymmetric, leafless, split-prime with limiting probabilities e^{-1} , $e^{-1/2}$, $e^{-3/2}$; labeled counts $\sim e^{-2}(2n)!/(2^{n+2}n)$ Relative errors $O(1/n)$ for c_n , g_n , the laws above and the labeled counts; $c_n/g_n = 1 - \frac{1}{2n} + O(n^{-2})$; $d_{\text{TV}}((X_n, Y_n, Z_n), \text{Poisson}(1)^{\otimes 3}) \leq \min\{1, 200/n\}$ for $n \geq 6$ Smooth inverse $\phi(x) = u + 1 + \frac{3+\log 2}{2 \log(2u)} + O(\frac{1}{u \log u})$; least-index brackets of width $O(1/(u \log u))$ inside two ceilings $c_{13} = 21,593,488,017$, derived by Euler inversion $\mathbb{E}[(1+x)^X(1+y)^Y(1+z)^Z] = e^{x+y+z}(1 + F/M + O(M^{-2}))$ uniformly on compacts; $\mathbb{P}(X = Y = Z = 0) = e^{-3}(1 - 10/M + O(M^{-2}))$ Three-chord interval lemma; $\mathbb{P}(\text{indecomposable}) = e^{-3}(1 - 5/n + O(n^{-2}))$; $p_n^a = e^{-3} h_n(1 - 5/n + O(n^{-2}))$ Finite canonical split-tree transfer; rooted counts $b_0, \dots, b_3 = 1, 3, 11, 58$ $c_n = e^{-3/2} h_n(1 - \frac{15}{4n} + O(n^{-2}))$, $g_n = e^{-3/2} h_n(1 - \frac{13}{4n} + O(n^{-2}))$; in the Stirling normalization $-\frac{91}{24}, -\frac{79}{24}$ $c_n/g_n = 1 - \frac{1}{2n} - \frac{1}{n^2} + O(n^{-3})$; the transfer to every fixed order, <i>conditional</i> on the prime expansion to that order; conditional second coefficients $\beta - \frac{277}{32}, \beta - \frac{297}{32}$ Automorphism-weighted counts $c_n(t)$, $g_n(t)$; labeled counts with $-\frac{47}{12n}, -\frac{41}{12n}$; $\mathbb{P}(\text{Aut}(G) = 1) = e^{-1}(1 - \frac{1}{2n} + O(n^{-2}))$; $\mathbb{E} \text{Aut}(G) ^{-t}$; $\mathbb{P}(\text{Aut}(G) \text{ nonabelian}) = \frac{1}{2n} + O(n^{-2})$ Refined inverse with centres s_a , $K_c = \frac{163}{24}$, $K_g = \frac{91}{24}$, width $O(1/(u^2 \log u))$	Part I, Theorem 9.1, Lemma 9.2, (9.11) Part I, Propositions 10.1, 10.2, (10.14) Part I, Section 11 Part II, Theorem 16.1 Part II, Lemma 18.1, (18.3), (14.1) Part II, Theorem 19.1, Subsection 19.4 Part II, Theorem 14.1 Part II, Theorem 14.2, Section 21 Part II, Section 22 Part II, Theorem 23.1
<i>Open in both Parts</i> The prime second coefficient β , hence the second graph coefficients; an unconditional expansion to every fixed order; effective constants, onsets and certified thresholds; reconciliation with the 2026 journal text of the main source; an audit of the labeled consequences; unbounded automorphism observables. See Section 25.	

The two companions check finite identities, enumerations, normalizations and coefficient algebra in exact arithmetic; no asymptotic statement rests on them. The leading inverses of both Parts are instances of the transseries volume’s factorial core; their two-ceiling brackets are not instances of its staircase theorem (next section).

The OEIS entries, and the inverses

The entries as read by the write on 5 October 2026 are A156808 revision #9 (June 29, 2026) and A156809 revision #16 (June 30, 2026), the revisions of Part I's snapshot (shipped as `data/150-leading-companion-sources-oeis_snapshot.json`).

- **A156808** (Lars Eirik Danielsen, Feb 16 2009), “Number of nonisomorphic connected circle graphs of order n ”: terms for $n = 1, \dots, 12$, ending with 892278076; no formula, comment or asymptotic statement. Part I derives $c_{13} = 21,593,488,017$; the value is not in the entry, and nothing was submitted to the OEIS.
- **A156809** (Lars Eirik Danielsen, Feb 16 2009), “Number of nonisomorphic circle graphs of order n ”: terms for $n = 1, \dots, 13$; the extension line credits $a(13) = 22576188846$ to Tom Johnston, Oct 17 2020, whose blog post “Enumerating circle graphs” (2020, read by the write) gives that count for all circle graphs of order 13 and no connected count. No formula, comment or asymptotic statement.

Both entries cite Danielsen and Parker [2], whose Table 3 gives the counts through order 12. The theorems of this report are additions to the entries, not corrections of them.

The inverses. Both Parts write the threshold through u , the solution of $u(\log(2u) - 1) = L$ (Part I's (10.4), Part II's (23.1)). This is the factorial core `p0:prop:factorial-core` of the transseries volume [7], $X(\kappa \log X + d) = L$, with $\kappa = 1$ and $d = \log 2 - 1$; its solution $\exp(W_0(Le^d) - d)$ equals $L/W_0(2L/e) = u$, and its side condition $\log(2u) > 1$ holds for every $L > 0$ (it is $W_0(2L/e) > 0$). (Wording corrected after the independent check of 5 October 2026; the first wording read “holds for large L ”.) After taking logarithms it is the Lambert core `p0:thm:lambert-core` with $a = b = 1$, in the variable $\log(2u) - 1 = W_0(2L/e)$ and with right-hand side $\log L + \log 2 - 1$. The corrections to u (Part I's $1 + \frac{3+\log 2}{2\log(2u)}$, Part II's further $(K_a - B^2/(2D^2))/(uD)$) are derived in the Parts by Taylor expansion of the explicit Gamma-function model $H(t)$ at u ; they are not taken from the volume, and we do not cite them as instances of its centred expansions. The two-ceiling brackets (10.9), (10.14) and (23.2) follow the separation pattern of `p0:thm:staircase(2)` but are not instances of it: that theorem inverts an admissible interpolation of the sequence itself, while the Parts compare a_n with the model H , which does not interpolate a_n ($H(n) = e^{-3/2}h_n \neq a_n$), and prove the brackets directly at the integers. Neither manuscript cites the volume.

Notation across the two Parts

Common to both Parts, with the same meaning: c_n, g_n (offsets one; $g_0 = 1, m_0 = 1$); $m_n = (2n-1)!!$; $h_n = m_n/(4n)$; $\mathcal{M}_n, G_D, R_n(G)$; S_n , the number of indexed matchings fixed by a nonidentity dihedral symmetry; p_n^a (Part I: asymmetric split-prime circle graphs; Part II: “abstractly asymmetric prime circle graphs”, the same class); $M = 2n$ in the local analysis; X, Y, Z (and Part I's X_n, Y_n, Z_n) the counts of chords at cyclic distance one, chords at distance two, and pairs of consecutive endpoints matched to another consecutive pair; the smooth model $H(t) = e^{-3/2}2^t\Gamma(t + \frac{1}{2})/(4\sqrt{\pi}t)$; $u = L/W_0(2L/e)$ and N_a , the least index. The symbols below change meaning between the Parts or inside one; each Part's text fixes its meaning there, and each Part opens with a short table of its own readings. No symbol was renamed.

Symbol	Meanings
c_r, c_n	$c_n = \text{A156808}(n)$ throughout. Part II, Section 16: c_r is the coefficient of u^r in the partition polynomial $H_M(u)$ (16.4). <i>Tempting false reading:</i> that c_r is not a circle-graph count. Part II also writes $c_n(t)$ for the automorphism-weighted count.
B	Part I: $B = \frac{3}{2}(1 + \log 2)$ in (10.2); $B_0, \dots, B_3$ the sums of Section 9. Part II: $B = \sum_{P \sim Q} w_P w_Q$, $B_1(s)$, B_* (Section 17); $B_d(z)$ the rooted-branch polynomial (Section 19); $B_1 = -a^2 - a/2 + 2$ (Subsection 21.4); $B = \frac{3+\log 2}{2}$ in (23.1). <i>False reading:</i> Part II's inverse constant B is Part I's $B - \log 2$, not Part I's B .
A, D	Part I: $A = \log(2u)$ (Section 10); D a matching, D_0, D^α ; $\mathcal{D}_{n;\ell,t,f}$ the decorated families. Part II: $D = \log(2u)$ (Part I's A); D_n a uniform matching; $\mathcal{D} = s d/ds$; $A(s)$ the leading polymer sum, $A = P e^{3/2}$, a Poisson(3/2) variable A , A_1, A_2 polynomials, $A_x = N_{S_x}(\rho_x)$.
q, ρ	Part I: $q_s = m_s m_{n-s} / m_n$ (4.2); $\rho_r = \prod_{j < r} (M - 2j - 1)^{-1}$ (9.2). Part II: $q_r = \prod_{j < r} (M - 2j - 1)^{-1}$ (16.5), which is Part I's ρ_r , not its q_s ; q an expansion order (Section 21); $\rho = 2^{-t}$ and ρ_x a root vertex (Sections 19, 22).
K	Part I: a truncation level, fixed in Section 5 and $\lceil \log n \rceil$ in Section 9. Part II: $K_c = \frac{103}{24}$, $K_g = \frac{91}{24}$ in the inverse centres.
L, T, F	Part I: the decoration types L, T, F , their counts L_n, T_n, F_n ; $L = \log x$; $F(t) = t(\log(2t) - 1)$ in the proof of Proposition 10.1. Part II: $F(x, y, z)$ the local polynomial (16.2); $F_\delta(t) = \log H(t) + \delta/t$; $L = \log y$; L_1, L_2 polynomials; T a split tree.
H	Part I: H a prime core graph (Section 6); $H(t)$ the smooth model (10.1). Part II: $H_M(u)$ the partition polynomial; H a prime centre (Section 19); the same $H(t)$.
$\alpha, \beta, \gamma, \eta$	Part I: α, β index local patterns (Section 9); $\eta > 0$ and $\eta(x)$ bracket errors (Section 10). Part II: α and β the first and second relative prime coefficients ($\alpha = -5$ proved, β unknown), α_t ; $\gamma = \alpha + 5/4$, $\eta = \gamma + 1/2$.
C	A constant in (3.6) and in (15.3) (the same role); Part I: C_1, C_2, C_* inverse constants. Part II: C also a constant in (18.1) and in the polymer bounds, C_* a remainder sum (Section 17), C_a, y_a inverse constants; $C_n^{\text{lab}}, G_n^{\text{lab}}$ the labeled counts, which Part I writes $c_n^{\text{lab}}, g_n^{\text{lab}}$.
S	S_n as above (both). Part I: S_α a pattern support. Part II: S a bound $ u \leq S/M$, S_P a polymer support, (S_x, ρ_x) a rooted branch, S_3 the symmetric group.
x, y, z	Part I: x the threshold input. Part II: x, y, z complex weights (Section 16); y the threshold input (Section 23).
d, k, r, s	Part I: d a rotation order and a root displacement; k a decomposition size or decoration count; r a deletion size or arc parameter, r_n a log error; $s = r + 1$, $s(x)$ the inverse centre. Part II: d a rotation order, a branch-size bound, d_n^k the decomposed-matching count, d_0, d_1 displacements; r a split-side bound, $r(A)$ a chord count, r_s rooted classes, $r_\pm$ roots; s a complex variable, a split side, $s_a(y)$ the inverse centres.
P, W	Part I: $P \sim \text{Poisson}(1)$ (Proposition 8.2); W_0 Lambert. Part II: P the prime amplitude (e^{-3}), P, Q polymers; W the polymer sum, $W_1(s)$, W_* ; W_0 Lambert.
b	Part II only: b_j the rooted connected circle-graph counts, $b_j(t)$ their weighted versions; $b = a_2$ an excess-two multiplicity (Subsection 21.4 and Section 20).

Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243. Their author lines read “Report 150” and “Report 152”; neither names an author, a tool or an addressee (each sets an empty PDF author field), neither carries “prepared for private review” wording, and neither pins a ProveIt commit or names a repository path. They arrived in commit `60f54ea06` and were placed by `47fc7a069` (batch 106).

- **Part I**, bundle Report 150, archive `Circle_Graph_Enumeration_Error_Bounds_and_Inverses_Source.zip` (622,124 bytes, 18 files; `Report150.tex`, 1445 lines, 23 pages, dated 3 October 2026); the base. Contributes the setting, the symmetry and deletion bounds, the fibre identity, the leaf lemmas, the upper bound by independent leaf moves, the lower bound by canonical recovery of decorations, the transfer to disconnected graphs, saturation with the core and automorphism laws and the labeled equivalent, the forcing coupling with the self-contained Poisson approximation and the quantitative sandwich, the smooth inverse with its brackets, and the finite checks with the derived c_{13} .
- **Part II**, bundle Report 152, archive `Circle_Graph_First_Corrections_and_Refined_Inverses_Source.zip` (630,050 bytes, 18 files; `Report152.tex`, 1200 lines, 19 pages, dated 3 October 2026). Contributes the uniform local expansion through the polymer representation, the global matching-probability remainder, the three-chord interval lemma and the first prime correction, the finite canonical split-tree transfer with its tails, the first graph coefficients, the second connectivity term, the conditional transfer to every fixed order and the conditional second coefficients, the automorphism-weighted and labeled corrections, and the refined inverse.

Where the merge had to choose:

1. *Order and base.* Base and dependency order agree: Report 150 is Part I, its sequel Report 152 Part II (see the Guide).
2. *Restatements.* Part II's Section 15 restates in its own words the structural inputs Part I takes from [1] and [4], including the faithfulness argument for prime chord actions, and re-proves Part I's symmetry bound (Lemma 3.1) and coarse bound (3.5)–(3.6), so that its proof does not depend on Part I. The two manuscripts share about 1.5% (of Report 150) and 1.8% (of Report 152) of their word 8-grams (intake measurement). The restatements are printed in full, with notes pointing to Part I.
3. *Supersession.* Part II's Theorem 14.1 identifies the first relative coefficients that Part I's Theorem 9.1 bounds; its Theorem 14.2 sharpens (1.5) and (9.1); its Theorem 23.1 refines Part I's Section 10; its Section 22 refines Part I's asymmetry probability and labeled equivalent. Part I's superseded statements stay, with dated notes. Only Part I has the $\text{Poisson}(1/2)^{\otimes 3}$ laws with total-variation rates, the explicit bound $200/n$, the leafless and split-prime probabilities, saturation, and c_{13} .
4. *Front matter and bibliography.* The two title blocks and abstracts are printed at the heads of their Parts, with each manuscript's scope paragraph; one table of contents replaces two. The two bibliographies are merged: the entries for [1], [4] and the two OEIS entries occur in both manuscripts with different annotations, and both annotations are kept; [2] is Part I's, [3] Part II's; the write added [7].
5. *Appendix.* Report 150's Appendix A ("Dependency map and reproducible release") is printed as Section 13, so that Part II's sections follow it.
6. *Write additions.* This front matter, the reading-conventions tables of the Parts, Remark 3.3, Section 25, the dated notes, labels for unlabelled sections, and the label prefixes. Everything else is delivered text.

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier 106-GRAPHS of batch 106, 5 October 2026) both manuscripts were read in full and no step was found false. Checked there: the Euler transform of A156808 gives A156809 through order 12; the derived c_{13} ; the constant $B = \frac{3}{2}(1 + \log 2)$ from Stirling's formula for $\Gamma(t + \frac{1}{2})$; that the upper and lower constants of Part I agree; $F(-1, -1, -1) = -10$; $\mathbb{E}[-A^2 + 3A/2] = -\frac{3}{2}$ for $A \sim \text{Poisson}(\frac{3}{2})$ and the arithmetic giving $-\frac{15}{4}$, $-\frac{13}{4}$, K_c , K_g , $-\frac{91}{24}$, the labeled values $b_1(1) = 2$, $b_2(1) = \frac{19}{3}$ and $\frac{13}{12}$; and the delivered suites, whose recorded outputs were regenerated byte for byte on copies (README). The intake also ran a Monte Carlo simulation of $\mathbb{P}(X = Y = Z = 0)$ (note at Theorem 16.1; uncertified).

The write read the cited statements of the external inputs: in arXiv:2402.06394v1 of Bassino, Bouvel, Féray, Gerin and Pierrot (the file whose SHA-256 both manuscripts record; arXiv lists no later version as of 5 October 2026) Definition 5.3, Proposition 5.5, Lemmas 5.6–5.10, Propositions 5.11, 5.12 and 5.14 and the bound (12); in Klavík and Zeman's STACS paper (the recorded file) Lemma 7 (Aut of a split tree equals Aut of its graph) and the proof sketch of Lemma 8 on p. 549, which says that a prime circle graph has a unique circle representation up to rotations and reflections, so that its automorphism group is a subgroup of the dihedral group; in Gioan and Paul's arXiv:0810.1823v2 Lemmas 2.3 and 2.8, Corollary 2.6, Theorem 2.9 and Corollary 2.10 (the published version was not compared); and Table 3 of Danielsen and Parker's arXiv:0804.2576v2. Each says what the Parts take from it. The write recomputed c_{13} and the ratios quoted in the notes to Section 11; by brute force over all graphs on two to five vertices, the rooted counts $b_0, \dots, b_3$ and the root-stabilizer orders used in Section 22; the sum $353.96 < 354$ behind (9.11); and, numerically, the transfer arithmetic of (21.4) (note there). Remark 3.3 is proved by the write.

What was *not* done: the analytic arguments (the Poisson approximation of Lemma 9.2, the polymer bounds and the hard-core remainder, the global matching remainder, the split-tree transfer and its tails, the inverse expansions) were read and their displayed algebra checked, but they are the manuscripts' proofs, corroborated by finite checks, not independently verified. Not read: the 2026 journal version of Bassino et al. (*Electronic Journal of Probability* **31**, article 110), at intake or at the write (its publisher page refused automated access); Gabor, Supowit and Hsu's paper, cited through [1, Proposition 5.12]; and, beyond its abstract, the paper of Arratia, Bollobás, Coppersmith and Sorkin. Priority is therefore bounded as both manuscripts bound it: their results were not located in the sources checked, which is not a claim of novelty against the final journal text.

Relation to neighbouring reports and formal projects

No other report of the collection treats circle graphs or A156808 and A156809 (repository search, 5 October 2026). Two reports placed in the same commit and written separately in batch 106 are siblings: `oeis-sequence-asymptotics/a123448-permutation-graphs` (bundle Report 153, unlabeled permutation graphs, OEIS A123448) and `oeis-sequence-asymptotics/a005975-interval-graphs` (bundle Report 133, unlabeled interval graphs, OEIS A005975 and A005976). The three share a pipeline: count a representation model (here indexed matchings; there permutations and interval orders), show that almost every graph has a recoverable prime core whose representations form one full orbit of the representation symmetry group (here dihedral of order $4n$; there the Klein four-group and a duality of order two), divide by that group's order, make the symmetric cores negligible, and invert with ceiling-safe Lambert- W brackets. They share no statement: the decomposition theories (split trees here, modular decomposition there), the external inputs

and the normalizations $((2n - 1)!!/(4n)$, $n!/4$, half the Fishburn numbers) differ, no manuscript cites another, and a merged report would give several symbols two to four meanings (placement commit 47fc7a069). Bassino et al. [1] treat permutation graphs as well. The multigraph reports **a007716-bipartite-multigraphs** and **a307316-leafless-multigraphs** use the same rare-symmetry philosophy for other objects and share no statement with this one. The leading inverses are instances of the transseries volume's factorial core (above). Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats circle graphs, and no statement of this report is formalized.

Part I

Leading asymptotics of unlabeled circle graphs

Bundle Report 150, archive `Circle_Graph_Enumeration_Error_Bounds_and_Inverses_Source.zip`, delivered as `Report150.tex` (1445 lines, 23 pages), the base of the merge. Delivered title block “Leading asymptotics of unlabeled circle graphs / Connected and all graphs in OEIS A156808 and A156809 / Report 150 / 3 October 2026”. Printed as delivered, with **[write]** notes; section, statement and equation numbers are those of the delivery, and its Appendix A is Section 13. Remark 3.3 is added in the write.

Reading conventions of Part I	Elsewhere in the report
$B = \frac{3}{2}(1 + \log 2)$ in (10.2); $B_0, \dots, B_3$ the sums of Section 9	Part II: $B = \frac{3+\log 2}{2}$ (this Part’s $B - \log 2$) in the inverse; B also a conflict sum and $B_d(z)$ a branch polynomial
$A = \log(2u)$ (Section 10)	Part II: $D = \log(2u)$; A an amplitude, a Poisson variable, polynomials
$q_s = m_s m_{n-s} / m_n$; ρ_r the probability of r prescribed chords	Part II: q_r is this Part’s ρ_r
K a truncation level	Part II: K_c, K_g inverse constants
L, T, F decoration types; $L = \log x$; H a core graph and $H(t)$ the model	Part II: $F(x, y, z), F_\delta$; $L = \log y$; $H_M(u)$
α, β pattern indices; $\eta, \eta(x)$ bracket errors	Part II: $\alpha = -5, \beta$ prime coefficients; $\eta = \gamma + 1/2$
$c_n^{\text{lab}}, g_n^{\text{lab}}$	Part II: $C_n^{\text{lab}}, G_n^{\text{lab}}$

Abstract of this Part (as delivered). Let c_n and g_n count connected and arbitrary unlabeled circle graphs on n vertices. We prove

$$c_n \sim g_n \sim e^{-3/2} \frac{(2n-1)!!}{4n}.$$

The proof combines published local Poisson limits and uniqueness of prime representations with two model-specific arguments: inverse representation-fiber weights give an upper bound, while canonically recoverable leaf and twin decorations of asymmetric prime cores give the matching lower bound. We also prove the connected fraction $1 - 1/(2n) + o(1/n)$, three independent limiting Poisson decoration counts of mean $1/2$, a Poisson core deficit of mean $3/2$, and asymmetry probability e^{-1} . A smooth Lambert- W inverse is converted into ceiling-safe least-index brackets. A self-contained forcing-coupling argument sharpens both individual count estimates to relative error $O(n^{-1})$ and the connectivity error to $O(n^{-2})$. Fresh exact finite checks and a deterministic source package accompany the proofs.

Scope. The complete primary texts inspected were arXiv:2402.06394v1 (2024), an additional complete author-hosted 2024 copy, and the references specified below. The 2026 journal version was identified bibliographically but its final text was not inspected. The result fills a leading-equivalent gap in the sources checked; this is not an exhaustive priority claim. The proof gives neither an all-orders expansion nor an effective finite- n onset. Finite checks are not substitutes for the structural proof. No external submission, author contact, or public repository change is part of this report.

[write] Scope, 5 October 2026. The bounded review stated here is kept. The write read arXiv:2402.06394v1 at the places this Part cites; arXiv lists no later version as of 5 October 2026. The journal text was read neither at intake nor at the write (“What was checked, and what was not” in the front matter). The first correction coefficients that this Part leaves open are identified in Part II.

1 The result and its counting conventions

A circle graph is the intersection graph of finitely many chords of a circle, with two vertices adjacent exactly when their corresponding chords cross. All graphs in this report are finite, simple and undirected. Chord endpoints can be assumed distinct. An unlabeled graph means an isomorphism class, counted once regardless of its automorphism group.

Let $\mathcal{M}_n$ be the perfect matchings of a fixed cyclically ordered endpoint set $\{0, 1, \dots, 2n-1\}$. Its cardinality is

$$m_n = (2n-1)!! = \frac{(2n)!}{2^n n!}, \quad m_0 = 1. \quad (1.1)$$

These are *indexed* matchings: the endpoints are indexed, but the chords do not have independent labels. Write G_D for the unlabeled intersection graph of $D \in \mathcal{M}_n$. The dihedral group acting on the endpoints has order $4n$. A matching is *geometrically asymmetric* when its stabilizer in that group is trivial. A graph is *asymmetric* when its graph automorphism group is trivial. These two notions must not be interchanged for general circle graphs.

Set

$$c_n = \text{A156808}(n), \quad g_n = \text{A156809}(n), \quad h_n = \frac{m_n}{4n} \quad (n \geq 1). \quad (1.2)$$

Both OEIS sequences begin at $n = 1$. We use $g_0 = 1$ for the empty graph when discussing disjoint unions.

Theorem 1.1 (Leading enumeration). *As $n \rightarrow \infty$,*

$$c_n \sim g_n \sim e^{-3/2} h_n \sim \frac{e^{-3/2}}{2\sqrt{2}n} \left(\frac{2n}{e}\right)^n. \quad (1.3)$$

Moreover,

$$g_n - c_n = g_{n-1} + O(m_n/n^3) = c_{n-1} + O(m_n/n^3), \quad (1.4)$$

$$\frac{c_n}{g_n} = 1 - \frac{1}{2n} + o(n^{-1}). \quad (1.5)$$

Section 9 strengthens the individual equivalents to relative error $O(n^{-1})$ and the error in (1.5) to $O(n^{-2})$. The individual first correction coefficients are not identified.

[write] **Answered in Part II, 5 October 2026.** Part II's Theorem 14.1 identifies the individual first corrections: $c_n = e^{-3/2} h_n (1 - \frac{15}{4n} + O(n^{-2}))$ and $g_n = e^{-3/2} h_n (1 - \frac{13}{4n} + O(n^{-2}))$. Its Theorem 14.2 sharpens (1.5) to $1 - \frac{1}{2n} - \frac{1}{n^2} + O(n^{-3})$. This theorem and Section 9 are printed as delivered.

The main asymptotic normalization is h_n , rather than m_n . Dividing all matchings by $4n$ alone does not count circle graphs: a graph may have several inequivalent chord representations, and geometric symmetries can shorten individual orbits. Sections 3–6 handle both issues explicitly.

2 Published inputs and the bounded source review

2.1 Splits and prime representations

A split of a graph is a partition $V = A \sqcup B$ whose crossing edges form $A' \times B'$ for some $A' \subseteq A$ and $B' \subseteq B$; the crossing set is allowed to be empty. The split is nontrivial if $|A|, |B| \geq 2$. A graph is *split-prime* if it has no nontrivial split.

Bassino, Bouvel, Féray, Gerin and Pierrot [1, Definition 5.3 and Proposition 5.5] define matching decompositions by four successive cyclic intervals, allowing empty intervals, such that every chord lies within one interval or joins opposite intervals. A k -decomposition uses the source's anchored convention: C_1 contains endpoint 0 (called 1 in the source), the other nonempty intervals follow in cyclic order, and k counts the chords in $C_2 \cup C_4$. Thus k and $n - k$ are not freely interchangeable in the cited probability bound. A decomposition is nontrivial when $2 \leq k \leq n - 2$. Their equivalence is

$$D \text{ is indecomposable} \iff G_D \text{ is split-prime.} \quad (2.1)$$

We accept this published structural theorem. We also use their Proposition 5.12, attributed there to Gabor, Supowit and Hsu: a split-prime circle graph of order at least five has one matching representation up to rotation and reflection.

For the automorphism statement we use the natural geometric action, not just an isomorphism between abstract groups. Klavík and Zeman [4, proof of Lemma 8 and the following geometric explanation, pp. 548–549] describe prime graph automorphisms as rotations or reflections of the unique circle representation. Thus each prime graph automorphism is induced by such a symmetry.

For completeness, no nonidentity geometric symmetry can fix every chord of a split-prime representation of order at least five. A rotation fixing every chord would have order two and all chords would be diameters, giving a complete graph. A reflection fixing every chord has all but at most one chord perpendicular to its axis; these chords do not cross one another. The possible chord on the axis gives at most a star together with isolated vertices. None of these graphs is split-prime at this order. The natural action on chords is therefore faithful. In particular,

$$\text{a prime matching is geometrically asymmetric} \iff \text{its graph is asymmetric.} \quad (2.2)$$

No such equivalence is asserted outside the prime class.

2.2 The local Poisson input

For a uniform $D_n \in \mathcal{M}_n$, let X_n count chords joining consecutive endpoints and Y_n count chords joining endpoints at cyclic distance two. Thus a chord counted by Y_n has exactly one endpoint in its shorter open arc, for all sufficiently large n . Let Z_n count unordered pairs of disjoint consecutive endpoint pairs whose endpoints are matched to one another; each such local configuration is counted once, with either the crossing or noncrossing pairing allowed. These definitions agree with [1, Section 5.2.4] in the asymptotic range.

The precise published inputs are

$$(X_n, Y_n, Z_n) \xrightarrow{d} (X, Y, Z), \quad (2.3)$$

$$X, Y, Z \text{ are independent Poisson}(1),$$

$$\mathbb{P}(D_n \text{ has a } k\text{-decomposition for some } k > 2) = O(n^{-1}), \quad (2.4)$$

$$\mathbb{P}(D_n \text{ has no 2-decomposition}) \longrightarrow e^{-3}. \quad (2.5)$$

These are respectively Lemma 5.10 (proved in Appendix A.2), Lemma 5.7, and Lemma 5.8 of [1]. Their Lemma 5.9 identifies the simultaneous vanishing of X_n, Y_n, Z_n with the absence of the two endpoint decomposition sizes. The union bound and (2.4)–(2.5) immediately give

$$\mathbb{P}(D_n \text{ is indecomposable}) = e^{-3} + o(1). \quad (2.6)$$

Although their Proposition 5.11 states only the eventual lower bound e^{-4} , this sharper limit follows directly from the preceding lemmas. It is not a new Poisson theorem proved here.

[write] **What the source states, 5 October 2026.** Checked in arXiv:2402.06394v1. Its Proposition 5.11 states that, for n large enough, the probability that a uniform matching of size n is indecomposable exceeds e^{-4} . Its proof bounds the probability of decomposability by that of a 2-decomposition plus that of a k -decomposition with $k > 2$, and Lemmas 5.7 and 5.8 make this sum tend to $1 - e^{-3}$. So the limit e^{-3} in (2.6) follows from the source’s own lemmas (the upper bound from Lemma 5.8 alone), as this Part says: it sharpens a stated bound and corrects nothing in the source. The source uses e^{-4} for its lower bound (12), that the number of circle graphs of order n is at least $e^{-4}m_n/(4n)$ for large n ; Theorem 1.1 replaces that bound by an equivalent. Part II sharpens (2.6) to $e^{-3}(1 - 5/n + O(n^{-2}))$ in (18.3).

2.3 What was and was not inspected

The version actually used is arXiv:2402.06394v1, dated 9 February 2024. A complete author-hosted copy with PDF metadata dated 13 February 2024 was also checked; its relevant text agrees with that version and still gives the coarse bound. Journal metadata lists *Electronic Journal of Probability* **31** (2026), article 110, DOI [10.1214/26-EJP1570](https://doi.org/10.1214/26-EJP1570). The final journal text was not obtained and is not being cited as inspected.

Danielsen and Parker [2, Section 3, Table 3] supply exact circle-graph counts through order twelve. The inspected version is arXiv:0804.2576v2 (1 October 2009). The OEIS entries [5, 6] provide the same connected data and one additional all-graph value at order thirteen. Neither inspected OEIS entry states the equivalent (1.3). An older OEIS-linked paper of Arratia, Bollobás, Coppersmith and Sorkin was checked only at abstract level; its Euler-circuit enumeration is not used as a theorem about graph-isomorphism classes.

The literature claim is deliberately bounded: the equivalent was not located in these inspected texts or the accompanying targeted catalogue search. This is neither proof of universal novelty nor a claim to settle a well-known open problem. A final-version source reconciliation remains useful future work.

[write] **The sources as read at the write, 5 October 2026.** Table 3 of arXiv:0804.2576v2 (p. 8) agrees with Table 1 through order 12 and with the shipped extraction `data/150-leading-companion-sources-danielsen_parker_table3.txt`, which also carries the table’s two columns of local-complementation orbit counts. The live OEIS entries are the revisions of the shipped snapshot (A156808 #9, A156809 #16); the value $g_{13} = 22576188846$ is credited there to Tom Johnston (Oct 17 2020), whose blog post “Enumerating circle graphs” (2020) gives the all-graph count at order 13 only. Neither entry has a formula or a comment. The “coarse bound” above is the source’s Proposition 5.11, the bound e^{-4} (shipped provenance record `data/150-leading-SOURCE_PROVENANCE.json`), which the source turns into its lower bound (12) on the number of circle graphs (note after (2.6)). Remark 3.3 relates the source’s Proposition 5.14 to Lemma 3.1.

3 Symmetry bounds and the exact fiber identity

3.1 Every geometric exception is negligible

Let S_n be the number of indexed matchings fixed by some nonidentity symmetry of the regular $2n$ -gon.

Lemma 3.1. *For every fixed $A > 0$, $S_n = o(m_n n^{-A})$. In particular, $nS_n/m_n \rightarrow 0$.*

Proof. A rotation of order $d \geq 2$ has $k = 2n/d$ endpoint cycles. If a matching is invariant, it pairs two distinct cycles using one of d offsets, or pairs a cycle internally. The latter is possible only when d is even and must pair opposite points. Hence its number of fixed matchings is

$$k! [z^k] \exp \left(\mathbf{1}_{2|d} z + \frac{d}{2} z^2 \right). \quad (3.1)$$

For a power series with nonnegative coefficients, $[z^k]F(z) \leq F(r)r^{-k}$. Choosing $r = \sqrt{k/d}$ and using $k! \leq k^k$, the quantity in (3.1) is at most

$$\begin{aligned} (kd)^{k/2} \exp(k/2 + \sqrt{k/d}) &\leq (2en)^{n/d} \exp(\sqrt{n}) \\ &\leq (2en)^{n/2} \exp(\sqrt{n}). \end{aligned} \quad (3.2)$$

There are two reflection types. A reflection through gaps has no fixed endpoints, and its fixed-matching count is exactly the $d = 2$, $k = n$ instance of (3.1). A reflection through opposite endpoints has two fixed endpoints. They must be paired to each other: a fixed endpoint paired to a nonfixed one would acquire two partners under reflection. On the remaining endpoints the count is the $d = 2$, $k = n - 1$ instance. Both satisfy (3.2). A union bound over fewer than $4n$ nonidentity symmetries gives

$$S_n \leq 4n(2en)^{n/2} \exp(\sqrt{n}). \quad (3.3)$$

Stirling's formula gives $\log m_n = n \log(2n) - n + O(1)$, so the logarithm of the right side of (3.3) divided by m_n is $-\frac{1}{2}n \log n + O(n)$. This is smaller than $-A \log n$ for each fixed A and large enough n . $\square$

This argument explicitly covers both reflection types. It also bounds every symmetric matching, a stronger form of exception control than bounding only graphs with fewer than $4n$ representatives.

3.2 Representation fibers

For an unlabeled circle graph G of order n , define

$$R_n(G) = \#\{D \in \mathcal{M}_n : G_D \cong G\}.$$

This counts all indexed matching representatives, including all inequivalent dihedral orbits. Grouping the sum by its graph gives the exact identities

$$g_n = \sum_{D \in \mathcal{M}_n} \frac{1}{R_n(G_D)}, \quad c_n = \sum_{D \in \mathcal{M}_n} \frac{\mathbf{1}_{G_D \text{ connected}}}{R_n(G_D)}. \quad (3.4)$$

A geometrically asymmetric matching has an orbit of size $4n$; therefore $R_n(G_D) \geq 4n$ for such a matching. On symmetric matchings we may use the crude bound $1/R_n(G_D) \leq 1$. Consequently

$$g_n \leq h_n + S_n = O(m_n/n). \quad (3.5)$$

Fix once and for all a constant C such that

$$g_j \leq C m_j / j \quad (j \geq 1), \quad (3.6)$$

enlarging C to handle the finite initial segment. No explicit value of this constant or its initial asymptotic onset is required.

3.3 Deleting a bounded set of chords

Lemma 3.2. *For fixed K and $A > 0$, the probability that some deletion of at most K chords from D_n leaves a geometrically symmetric matching is $o(n^{-A})$.*

Proof. For a fixed deletion size $r \leq K$, choose its $2r$ endpoints, match them in m_r ways, and choose a symmetric matching on the remaining endpoints, with their cyclic order compressed. There are at most

$$\binom{2n}{2r} m_r S_{n-r} \quad (3.7)$$

such choices. Multiple counting only increases this bound. For fixed r , the ratio m_{n-r}/m_n is $O(n^{-r})$, while $\binom{2n}{2r} = O(n^{2r})$. Lemma 3.1, with a sufficiently large exponent, makes (3.7) divided by m_n smaller than any prescribed power of n^{-1} . Sum over $0 \leq r \leq K$. $\square$

This is uniform over the choice of deleted chords. Thus the chords can later be chosen from the graph's leaf set without any unjustified conditioning argument.

Remark 3.3 ([**write**] Credit for the method of Lemma 3.1, 5 October 2026). The method of Lemma 3.1, the exponential generating function (3.1) of the matchings fixed by a rotation of order d and a Cauchy bound on the circle of radius $\sqrt{k/d}$, is that of [1, Proposition 5.14]: the proportion of matchings of size n whose circle graph has fewer than $4n$ representatives is $o(n^{-n/3})$, proved by bounding the matchings fixed by a nonidentity symmetry. Neither Part cites that proposition. Its proof says that a reflection and a rotation of order two give the same enumeration sequence. That is exact for the n reflections whose axis passes through gaps between endpoints. (These are the source's reversals; its relabelling, read as exchanging $2n+1-i$ and $n+i$ for $1 \leq i \leq n/2$, maps the matchings fixed by the half-turn onto those fixed by the reversal $i \mapsto 2n+1-i$; as printed, with $2n-i$, it is off by one. Added after the independent check of 5 October 2026.) A reflection through two opposite endpoints fixes t_{n-1} matchings, by the argument of Lemma 3.1, where $t_k = k! \lfloor z^k \rfloor \exp(z+z^2)$ is the case $d=2$ of (3.1); a rotation of order two fixes t_n . The source's bound is unaffected, because $t_{n-1} < t_n$ for $n \geq 2$.

Proof of $t_{n-1} < t_n$. Put $f(z) = \exp(z+z^2) = \sum_k a_k z^k$, so $t_k = k! a_k$. From $f' = (1+2z)f$, $(k+1)a_{k+1} = a_k + 2a_{k-1}$ for $k \geq 1$, that is $t_{k+1} = t_k + 2k t_{k-1}$. With $t_0 = t_1 = 1$ every t_k is positive, so $t_{k+1} > t_k$ for $k \geq 1$. $\square$

Lemma 3.1 treats the two reflection types separately. Its conclusion $S_n = o(m_n n^{-A})$ is weaker than the source's $o(n^{-n/3})$ proportion, but (3.3) is explicit, and it suffices for both Parts.

4 Leaves of a typical indexed matching

A chord counted by Y_n is called *short*. Its shorter arc contains one endpoint, so it crosses exactly the chord incident to that endpoint. It is therefore a graph leaf. Call its unique neighbor its parent.

Lemma 4.1. *The expected number of graph leaves which are not short is $O(n^{-1})$.*

Proof. A graph leaf crosses exactly one other chord. The smaller open arc between its endpoints must contain an odd number $2r+1$ of endpoints: one is matched outside and the other $2r$ are matched internally. For a nonshort leaf $r \geq 1$. After its two endpoints are fixed, choose the inside

and outside endpoints of the unique crossing chord. All remaining matchings lie independently inside or outside the arc, so the exact number of completions is

$$\begin{aligned} (2r+1)(2n-2r-3)m_r m_{n-r-2} \\ = m_{r+1} m_{n-r-1}. \end{aligned} \quad (4.1)$$

For any specified arc length there are at most $2n$ possible leaf chords. Diameter choices, when present, are merely counted twice. Put $s = r + 1$ and

$$q_s = \frac{m_s m_{n-s}}{m_n}. \quad (4.2)$$

The expectation in question is at most

$$2n \sum_{s=2}^{\lfloor n/2 \rfloor} q_s. \quad (4.3)$$

For $0 \leq s \leq (n-1)/2$,

$$\frac{q_{s+1}}{q_s} = \frac{2s+1}{2n-2s-1} \leq 1. \quad (4.4)$$

Thus, for all sufficiently large n ,

$$\sum_{s=2}^{\lfloor n/2 \rfloor} q_s \leq q_2 + n q_3 = O(n^{-2}), \quad (4.5)$$

$$\begin{aligned} q_2 &= \frac{3}{(2n-1)(2n-3)}, \\ q_3 &= \frac{15}{(2n-1)(2n-3)(2n-5)}. \end{aligned}$$

Substitution into (4.3) proves the claim. $\square$

Lemma 4.2. *The probability that two short graph leaves have the same parent is $O(n^{-1})$.*

Proof. Choose the two short-arc positions in $O(n^2)$ ways. When their configuration is feasible, the two leaf chords and their common parent are three specified distinct matching edges on six endpoints: the parent joins the unique endpoint enclosed by each short leaf. A uniform matching contains these three edges with probability $m_{n-3}/m_n = O(n^{-3})$. Impossible or overlapping configurations contribute zero. Summing bounds the expected number of such pairs, hence also the probability, by $O(n^{-1})$. $\square$

In a connected graph of order greater than two, a leaf's parent cannot itself be a leaf. Lemmas 4.1 and 4.2 imply that, outside an unconditional event of probability $O(n^{-1})$, every graph leaf is short and all parents are distinct. The estimates remain adequate after we restrict a weighted sum to connected graphs: no conditional probability claim is needed.

5 An upper bound from independent leaf moves

Fix $K \geq 0$. Call a connected matching good if it has at most K graph leaves, all of them are short, their parents are distinct, and deleting all leaves leaves a geometrically asymmetric matching D_0 . By the previous lemmas, the bad connected matchings are contained in $\{Y_n > K\}$ and an event of probability $o(1)$. The geometric-symmetry contribution will be estimated separately.

Lemma 5.1 (Leaf moves). *If a good matching has k leaves, then its graph G satisfies*

$$R_n(G) \geq 4n 2^k. \quad (5.1)$$

Proof. View D_0 as a cyclic double-occurrence word, one symbol for each retained chord. At either occurrence of each parent symbol p , replace that occurrence by the block ℓ, p, ℓ , where ℓ is its deleted leaf. This new chord crosses just p . There are two choices per parent. Distinct parents have distinct symbols, so their endpoint occurrences are distinct and the local insertions commute, even when occurrences are consecutive. Every one of the 2^k choices represents G .

Suppose two expanded diagrams are dihedrally equivalent. The induced graph isomorphism preserves the full set of degree-one vertices, which is exactly the inserted leaf set. Delete that set on both sides. The remaining equivalence is a dihedral symmetry of D_0 with its cyclic order compressed. Its geometric asymmetry forces the identity on retained endpoints. The parent endpoints are therefore individually fixed. Because different leaves have different parents, each leaf must occupy the same chosen parent endpoint in both expansions. The choices agree. Hence the 2^k expanded diagrams lie in distinct dihedral orbits.

The same argument applied to a symmetry of a single expanded diagram gives the identity on all retained endpoints. For $n - K$ large enough there are at least three retained endpoints; an endpoint rotation or reflection fixing these individually is the identity. Therefore every expanded diagram is itself geometrically asymmetric and has exactly $4n$ indexed representatives in its orbit. Their disjoint orbits prove (5.1). $\square$

The lemma only gives a lower bound on the fiber size. It does not need to classify additional representations; extra representations would make the reciprocal weight still smaller.

For a connected matching with $n > 1$, necessarily $X_n = 0$, since a chord joining consecutive endpoints is an isolated graph vertex. On good matchings $k = Y_n$. Using (3.4), (5.1), the weight $1/(4n)$ on all asymmetric exceptions and the weight at most one on symmetric exceptions, we obtain

$$\frac{c_n}{h_n} \leq \mathbb{E}[\mathbf{1}_{X_n=0} \mathbf{1}_{Y_n \leq K} 2^{-Y_n}] + \mathbb{P}(Y_n > K) + o(1) + \frac{4nS_n}{m_n}. \quad (5.2)$$

The last term tends to zero by Lemma 3.1. Convergence in distribution of integer-valued variables gives convergence of each fixed finite collection of point probabilities, and the Poisson limit is tight. Thus (2.3) implies, for fixed K ,

$$\limsup_{n \rightarrow \infty} \frac{c_n}{h_n} \leq e^{-1} \sum_{j=0}^K 2^{-j} \frac{e^{-1}}{j!} + \mathbb{P}(\text{Poisson}(1) > K). \quad (5.3)$$

Letting $K \rightarrow \infty$ gives

$$\limsup_{n \rightarrow \infty} \frac{c_n}{h_n} \leq e^{-3/2}. \quad (5.4)$$

6 A lower bound from recoverable prime decorations

6.1 Asymmetric prime cores

Let p_N^a count asymmetric split-prime circle graphs of order N . By (2.6), the number of prime indexed matchings is $(e^{-3} + o(1))m_N$. Removing all geometrically symmetric matchings changes

this by at most $S_N = o(m_N)$. Prime uniqueness and (2.2) identify every remaining graph with one full orbit of size $4N$. Therefore

$$p_N^a \sim e^{-3 \frac{m_N}{4N}}. \quad (6.1)$$

A split-prime graph of order at least five is connected, has minimum degree at least two, and has no twin pair. Here vertices u, v are twins if their adjacencies to every vertex outside $\{u, v\}$ agree; they are true twins if adjacent and false twins if nonadjacent. A twin pair is a split side of size two. A leaf and its parent also form a split side of size two. Disconnected graphs admit an empty crossing split, or an equivalent split using an isolated vertex and another vertex when a component has order one. These observations establish the stated properties.

6.2 Three single-vertex decorations

Start from an asymmetric split-prime core H of order $N \geq 5$. Choose k distinct core vertices. At each selected vertex perform exactly one of these operations, each adding one vertex:

- L.* Add one pendant leaf adjacent to that vertex.
- T.* Replace the vertex by two adjacent twins, each with its former neighbors.
- F.* Replace the vertex by two nonadjacent twins, each with its former neighbors.

No core vertex receives more than one decoration. The resulting graph is connected and has $n = N + k$ vertices. All operations preserve the circle-graph class. For *L*, insert a chord around one endpoint as in Lemma 5.1. For *T* or *F*, duplicate the two endpoints of the core chord in very small adjacent neighborhoods. Alternating the two chord symbols makes them cross; nesting them makes them noncrossing. Both new chords meet exactly the former external neighbors. These local substitutions can be made simultaneously at distinct core chords.

Lemma 6.1 (Canonical recovery). *The decorated graph determines its core isomorphism class, every selected core vertex, and its L, T or F type. Two decoration patterns on an asymmetric core are isomorphic only if they are identical.*

Proof. Every old or cloned vertex retains at least the core's minimum degree two. The added *L* vertices are consequently exactly the degree-one vertices of the result. Delete precisely those leaves and save their incidences with their parents.

What remains is a blowup of the twin-free graph H with each replacement block of size one or two. Distinct core vertices u, v have a third vertex w whose adjacency distinguishes them, since otherwise u, v would be twins. Every replacement block for w is nonempty. Any vertex in that block continues to distinguish a vertex in the u block from a vertex in the v block. Thus twins in the blowup cannot lie in different blocks. The only nonsingleton maximal twin classes are exactly the intended size-two *T* and *F* blocks.

Contract each such class to one vertex. Its internal edge or nonedge records whether its type was *T* or *F*. The quotient is H , and the saved pendant incidences identify its *L* vertices. All steps are canonical under graph isomorphisms. An isomorphism between two outputs therefore induces an isomorphism between cores preserving all decorations. On the same asymmetric core that isomorphism must fix every vertex. This proves the assertion. $\square$

Twin swaps in the output graph are allowed, and indeed occur. There is no extra division by two for each twin pair: the count is of decoration patterns on the recovered asymmetric core, not of independently labeled clones.

6.3 Normalization and the lower sandwich

For fixed k the number of constructed graphs of order n is exactly

$$3^k \binom{n-k}{k} p_{n-k}^a \quad (6.2)$$

when $n-k \geq \max(5, k)$. The families for different k are disjoint, since recovery determines the core order. For every fixed K and all sufficiently large n ,

$$c_n \geq \sum_{k=0}^K 3^k \binom{n-k}{k} p_{n-k}^a. \quad (6.3)$$

For fixed k ,

$$\frac{m_{n-k}}{m_n} = \prod_{j=0}^{k-1} \frac{1}{2n-2j-1} \sim (2n)^{-k}, \quad \binom{n-k}{k} \sim \frac{n^k}{k!}. \quad (6.4)$$

Using (6.1), the exact normalization is

$$\begin{aligned} \frac{3^k \binom{n-k}{k} p_{n-k}^a}{h_n} &\sim e^{-3} 3^k \binom{n-k}{k} \frac{n}{n-k} \frac{m_{n-k}}{m_n} \\ &\rightarrow e^{-3} \frac{(3/2)^k}{k!}. \end{aligned} \quad (6.5)$$

First take the liminf in (6.3) with K fixed, and then increase K . This gives

$$\liminf_{n \rightarrow \infty} \frac{c_n}{h_n} \geq e^{-3} \sum_{k=0}^{\infty} \frac{(3/2)^k}{k!} = e^{-3/2}. \quad (6.6)$$

Together with (5.4), this proves the connected equivalent. The argument never assumes a uniform estimate when k grows with n and never interchanges an unbounded sum with an unproved asymptotic expansion.

7 Disconnected graphs and the first connectivity correction

Circle graphs are closed under disjoint union, by placing their chord diagrams in disjoint small arcs. Every disconnected graph has a component of order $k \leq n/2$. Choosing one such component and its complement may overcount, so

$$0 \leq g_n - c_n \leq \sum_{k=1}^{\lfloor n/2 \rfloor} g_k g_{n-k}. \quad (7.1)$$

No binomial coefficient occurs: these are unlabeled graph isomorphism classes, and disjoint union is a well-defined map from pairs of such classes.

Use (3.6) and $n - k \geq n/2$ to get

$$g_n - c_n \leq \frac{2C^2 m_n}{n} \sum_{k=1}^{\lfloor n/2 \rfloor} \frac{q_k}{k} = O(m_n/n^2), \quad (7.2)$$

where q_k is (4.2), $q_1 = 1/(2n - 1)$ and (4.5) bounds the remaining sum. Since $c_n \sim e^{-3/2} m_n/(4n)$, this proves $g_n/c_n = 1 + O(1/n)$ and hence the all-graph equivalent in Theorem 1.1. Stirling's formula proves its elementary factorial form.

For a sharper comparison, graphs with at least one isolated vertex are in bijection with all graphs on $n - 1$ vertices, by removing or adding one indistinguishable isolated vertex. The image is unique even when several isolates are present. Disconnected graphs with no isolated vertex have a component of order between 2 and $n/2$, and their count is at most

$$\sum_{k=2}^{\lfloor n/2 \rfloor} g_k g_{n-k} = O(m_n/n^3). \quad (7.3)$$

Thus $g_n - c_n = g_{n-1} + O(m_n/n^3)$. Also (7.2) at $n - 1$ gives $g_{n-1} - c_{n-1} = O(m_n/n^3)$. Finally,

$$\frac{g_{n-1}}{g_n} \sim \frac{h_{n-1}}{h_n} = \frac{n}{(n-1)(2n-1)} \sim \frac{1}{2n}. \quad (7.4)$$

Equations (1.4) and (1.5) follow.

8 Saturation and the typical recovered core

The upper and lower bounds agree exactly in the limit. This forces the recoverable decorated families to carry almost all unlabeled graphs; it does not require a structural classification of every exceptional graph.

For nonnegative integers ℓ, t, f , put $k = \ell + t + f$ and let $\mathcal{D}_{n;\ell,t,f}$ consist of constructions with exactly ℓ leaves, t true-twin pairs and f false-twin pairs, on distinct vertices of an asymmetric prime core of order $N = n - k$. Canonical recovery makes all these families disjoint, and for $N \geq \max(5, k)$ their exact size is

$$|\mathcal{D}_{n;\ell,t,f}| = p_N^a \frac{N!}{\ell!t!f!(N-k)!}. \quad (8.1)$$

Consequently, for every fixed triple,

$$\frac{|\mathcal{D}_{n;\ell,t,f}|}{h_n} \longrightarrow e^{-3} \frac{2^{-k}}{\ell!t!f!}. \quad (8.2)$$

Divide by either connected or all-graph counts, using Theorem 1.1, to get the limiting probability

$$e^{-3/2} \frac{(1/2)^\ell}{\ell!} \frac{(1/2)^t}{t!} \frac{(1/2)^f}{f!}. \quad (8.3)$$

The sum of (8.3) over all triples is one. Given $\varepsilon > 0$, choose a finite set of triples whose limiting mass exceeds $1 - \varepsilon$. Convergence on that finite set proves that the probability of the union of *all* decorated families has $\liminf$ at least $1 - \varepsilon$. Let ε decrease to zero. This proves saturation without exchanging an infinite sum and an asymptotic estimate.

Theorem 8.1 (Typical core and decoration laws). *For a uniform unlabeled circle graph of order n , either conditioned to be connected or not, with probability tending to one there is a canonically recovered asymmetric split-prime core of the form described in Lemma 6.1. Its decoration counts satisfy*

$$(L_n, T_n, F_n) \xrightarrow{d} (\text{Poisson}(1/2), \text{Poisson}(1/2), \text{Poisson}(1/2)), \quad (8.4)$$

with independent coordinates. Its deficit $n - |H|$ converges in distribution to $\text{Poisson}(3/2)$.

Proof. On the saturated family the counts and the core are uniquely recovered, and (8.3) gives their joint point probabilities. The finite-set argument also proves tightness, which yields the joint convergence. Outside this family define the counts arbitrarily; its probability tends to zero and cannot change the limit. Summing the three independent Poisson limits gives the deficit law. $\square$

The core claim concerns the explicit leaf-deletion and maximal-twin-contraction procedure on a probability-one limiting family. It is not an assertion that arbitrary iterative deletion and contraction orders give the same core for every graph.

8.1 Automorphisms and a labeled consequence

Proposition 8.2. *For a graph in $\mathcal{D}_{n;\ell,t,f}$,*

$$\text{Aut}(G) \cong (C_2)^{t+f}. \quad (8.5)$$

Thus a uniform unlabeled graph in either model is asymmetric with probability tending to e^{-1} , and its automorphism-group order converges in distribution to 2^P , where $P \sim \text{Poisson}(1)$.

Proof. Every automorphism preserves the leaves, the recovered twin classes, and the recovered core with its decorations. The core is asymmetric, so each core vertex is fixed. A leaf then is fixed by its distinct fixed parent. Each twin pair can be swapped independently; these swaps are automorphisms and are the only remaining freedoms. This proves (8.5). Saturation and Theorem 8.1 give $T_n + F_n \rightarrow \text{Poisson}(1)$ and the asserted consequences. The arbitrary exceptional graphs have vanishing probability. $\square$

In particular, geometric asymmetry of typical representations must not be used to claim asymmetry of typical graphs. The twin swaps above generally do not act as rotations or reflections of a given nonprime chord representation.

Other direct consequences are

$$\mathbb{P}(G \text{ leafless}) \longrightarrow e^{-1/2}, \quad \mathbb{P}(G \text{ split-prime}) \longrightarrow e^{-3/2}. \quad (8.6)$$

The first follows from $L_n = 0$. For the second, every nonempty decoration creates a nontrivial split: use a twin pair, or a leaf-parent pair, as a side of size two. The zero-decoration family is prime. Saturation accounts for all remaining cases.

Let g_n^{lab} and c_n^{lab} count graphs on a fixed labeled n -element vertex set. The number of labelings of an unlabeled graph is $n!/|\text{Aut}(G)|$. Hence

$$\frac{g_n^{\text{lab}}}{n!g_n} = \mathbb{E}_{\text{unlabeled}} \left[\frac{1}{|\text{Aut}(G)|} \right] \longrightarrow \mathbb{E}[2^{-P}] = e^{-1/2}. \quad (8.7)$$

The expectation passage is legitimate because the reciprocal automorphism size is bounded by one: first truncate to finitely many decoration triples, then use saturation to bound the remainder. No uniform integrability assertion about $|\text{Aut}(G)|$ is being made. The same argument holds in the connected model. Therefore

$$c_n^{\text{lab}} \sim g_n^{\text{lab}} \sim e^{-2} \frac{n!m_n}{4n} = e^{-2} \frac{(2n)!}{2^{n+2}n}. \quad (8.8)$$

This is a mathematical consequence of the proved unlabeled law. A separate literature and OEIS identification audit for labeled counts was not completed, so no separate priority claim is made.

[write] **Refined in Part II, 5 October 2026.** Part II's Section 22 gives the first corrections of these laws: $\mathbb{P}(\text{Aut}(G) = 1) = e^{-1}(1 - \frac{1}{2n} + O(n^{-2}))$ (22.5), $\mathbb{E}|\text{Aut}(G)|^{-1} = e^{-1/2}(1 - \frac{1}{6n} + O(n^{-2}))$, and the labeled counts (22.3)–(22.4) with relative corrections $-\frac{47}{12n}$ (connected) and $-\frac{41}{12n}$ (all). Part II's $C_n^{\text{lab}}, G_n^{\text{lab}}$ are this Part's $c_n^{\text{lab}}, g_n^{\text{lab}}$, and $n!e^{-2}h_n = e^{-2}(2n)!/(2^{n+2}n)$ is the leading term of (8.8). The leafless and split-prime laws (8.6) are this Part's alone. The labeled audit called for here was not made at the write either (Section 25).

9 Quantitative error bounds

The preceding proof establishes the leading equivalent using the published qualitative Poisson limit. We now prove a quantitative local approximation directly and propagate it through the same sandwich. This gives the following strengthening.

Theorem 9.1. *For $a_n = c_n$ or $a_n = g_n$,*

$$a_n = e^{-3/2}h_n(1 + O(n^{-1})), \quad \frac{c_n}{g_n} = 1 - \frac{1}{2n} + O(n^{-2}). \quad (9.1)$$

The exceptional probability outside the canonically decorated-core family is $O(n^{-1})$. Its three defect counts are within $O(n^{-1})$ total variation of independent Poisson(1/2) variables, and its core deficit is within $O(n^{-1})$ total variation of Poisson(3/2). The asymmetry, leafless and split-prime probabilities in Section 8 have additive error $O(n^{-1})$; the labeled equivalent (8.8) has relative error $O(n^{-1})$.

The constants and sufficiently large onsets for graph counts are not made explicit. Only the local-pattern bound below has a stated numerical constant. No individual first correction coefficient is identified.

[write] **Superseded in part, 5 October 2026.** For the counts, Part II's Theorem 14.1 identifies the first relative coefficients ($-\frac{15}{4}$ and $-\frac{13}{4}$ in the h_n normalization) and so implies the first estimate of (9.1); its Theorem 14.2 sharpens the second. The total-variation statements, the leafless and split-prime probabilities with error $O(n^{-1})$, and the explicit local bound (9.11) are this Part's alone.

Atomic patterns and an exact forcing coupling

Put $M = 2n \geq 12$, with endpoints modulo M . A prescribed matching pattern of r edges occurs with probability

$$\rho_r = \prod_{j=0}^{r-1} (M - 2j - 1)^{-1}. \quad (9.2)$$

Take as atomic X patterns the M edges $\{i, i + 1\}$, as Y patterns the M edges $\{i, i + 2\}$, and as Z patterns the two possible matchings between each unordered pair of disjoint cycle edges. There are $M(M - 3)/2$ pairs of disjoint cycle edges, hence $M(M - 3)$ two-edge Z patterns.

These two-edge patterns are distinct for $M \geq 6$. Otherwise the same four endpoints would admit two distinct perfect matchings by cycle edges; their symmetric difference would be a four-cycle inside C_M , which is impossible. Summing pattern indicators therefore gives exactly the source variables X_n, Y_n, Z_n . Their means are all equal to

$$\lambda_X = \lambda_Y = M\rho_1 = \frac{M}{M-1}, \quad \lambda_Z = M(M-3)\rho_2 = \frac{M}{M-1} =: \lambda. \quad (9.3)$$

Let $\mathcal{I}$ be the finite pattern set, let I_α be the occurrence indicator of pattern α , and write $p_\alpha = \mathbb{E}I_\alpha$ and S_α for its endpoint support.

To force an edge $\{u, v\}$ into a matching, do nothing if it is already there. Otherwise replace $\{u, x\}, \{v, y\}$ by $\{u, v\}, \{x, y\}$. To force a two-edge pattern, apply this operation in a fixed order, excluding already forced endpoints on the next step. Denote the result by D^α .

The result has precisely the uniform law conditional on $I_\alpha = 1$. On L currently unforced endpoints, each matching containing a prescribed edge has $L - 1$ preimages: itself and two preimages for each of its $L/2 - 1$ other edges. One force thus pushes the uniform matching law to the uniform conditional law. Applying this fact successively proves the assertion for two edges.

Every original matching edge whose endpoints lie outside S_α survives every forcing step. Only edges incident with a forced endpoint are removed. Therefore

$$S_\alpha \cap S_\beta = \emptyset \implies I_\beta(D^\alpha) \geq I_\beta(D). \quad (9.4)$$

This is a statement about disjoint supports only.

A self-contained Poisson approximation bound

We use total variation in the convention $d_{\text{TV}}(P, Q) = \sup_B |P(B) - Q(B)|$.

Lemma 9.2 (Reduced conditional coupling bound). *Let $\Xi = (I_\alpha)_{\alpha \in \mathcal{I}}$ be any finite family of indicators with means p_α . Suppose Ξ^α has the law of $\Xi - \delta_\alpha$ conditional on $I_\alpha = 1$. If Π has independent Poisson coordinates of means p_α , then*

$$d_{\text{TV}}(\mathcal{L}(\Xi), \mathcal{L}(\Pi)) \leq \sum_\alpha p_\alpha \mathbb{E} \|\Xi - \Xi^\alpha\|_1. \quad (9.5)$$

Proof. For a function f on nonnegative integer vectors write $\Delta_\alpha f(\xi) = f(\xi + \delta_\alpha) - f(\xi)$. The independent immigration–death process has generator

$$\mathcal{A}f(\xi) = \sum_\alpha p_\alpha \Delta_\alpha f(\xi) - \sum_\alpha \xi_\alpha \Delta_\alpha f(\xi - \delta_\alpha).$$

Each initial particle dies at rate one; particles immigrate in coordinate α at rate p_α . Its equilibrium law is Π . For an event indicator J , the semigroup solution

$$f_J(\xi) = - \int_0^\infty (\mathbb{E}J(Z_\xi(t)) - \mathbb{E}J(\Pi)) dt$$

satisfies $\mathcal{A}f_J = J - \mathbb{E}J(\Pi)$. The integral is finite: couple equilibrium and nonequilibrium initial states with the same immigrants, so that their finite expected initial population difference survives with probability decaying as e^{-t} .

Couple processes with zero, one and two additional initial particles, including two in the same coordinate if needed. A mixed second difference of a function valued in $[0, 1]$ has absolute value at most two; it vanishes unless both additional particles survive. Every second difference of f_J is thus bounded in absolute value by $\int_0^\infty 2e^{-2t} dt = 1$. By telescoping coordinate changes, $\Delta_\alpha f_J$ is 1-Lipschitz in ℓ^1 . Conditional expectation of the death term now gives

$$\mathbb{E}\mathcal{A}f_J(\Xi) = \sum_{\alpha} p_{\alpha} \mathbb{E}[\Delta_{\alpha} f_J(\Xi) - \Delta_{\alpha} f_J(\Xi^{\alpha})].$$

Taking absolute values, then the supremum over events J , proves (9.5). $\square$

Use $\Xi_{\alpha}^{\alpha} = 0$ and $\Xi_{\beta}^{\alpha} = I_{\beta}(D^{\alpha})$ for $\beta \neq \alpha$. The coordinate α contributes p_{α}^2 to (9.5). By (9.4), a pair on disjoint supports contributes exactly

$$\mathbb{P}(I_{\alpha} = I_{\beta} = 1) - p_{\alpha}p_{\beta}.$$

An overlapping pair contributes at most $p_{\alpha}p_{\beta} + \mathbb{P}(I_{\alpha} = I_{\beta} = 1)$. We bound the diagonal, overlapping product, compatible overlap, and disjoint sums separately, always using ordered pairs when summing over α, β .

The diagonal sum is

$$B_0 = \sum_{\alpha} p_{\alpha}^2 = \frac{2M}{(M-1)^2} + \frac{M}{(M-1)^2(M-3)}. \quad (9.6)$$

At an endpoint v , cyclic symmetry and the support sizes give

$$\mu_v = \sum_{\alpha: v \in S_{\alpha}} p_{\alpha} = \frac{2\lambda_X + 2\lambda_Y + 4\lambda_Z}{M} = \frac{8}{M-1}.$$

Thus the product contribution from overlapping pairs is at most

$$B_1 \leq \sum_v \mu_v^2 = \frac{64M}{(M-1)^2}. \quad (9.7)$$

Including diagonal pairs in this upper bound is harmless.

Two compatible matching patterns sharing an endpoint must share its incident chord. Distinct singleton patterns cannot overlap compatibly. Every fixed chord belongs to at most four Z patterns: choose one cycle neighbor of each of its endpoints, after which the second chord is determined. There are $2M$ singleton patterns, so there are at most $16M$ ordered compatible singleton/ Z pairs. Each union forces two edges. Two distinct compatible Z patterns cannot share both chords, by the no-duplication argument. Choose their unique shared chord in at most $\binom{M}{2}$ ways and an ordered pair of distinct Z patterns containing it in at most $4 \cdot 3$ ways. Their union forces three edges. Therefore

$$\begin{aligned} B_2 &\leq 16M\rho_2 + 6M(M-1)\rho_3 \\ &= \frac{16M}{(M-1)(M-3)} + \frac{6M}{(M-3)(M-5)}. \end{aligned} \quad (9.8)$$

Incompatible pairs have zero joint occurrence probability.

For disjoint patterns with $r, s \in \{1, 2\}$ edges, the joint probability is ρ_{r+s} and

$$1 \leq \frac{\rho_{r+s}}{\rho_r \rho_s} = \prod_{j=0}^{s-1} \frac{M-2j-1}{M-2r-2j-1} \leq \frac{(M-1)(M-3)}{(M-5)(M-7)}.$$

The largest ratio occurs at $r = s = 2$. With $\delta_M = 8(M-4)/((M-5)(M-7))$, the total disjoint contribution is consequently bounded by

$$B_3 \leq \delta_M \left(\sum_{\alpha} p_{\alpha} \right)^2 = \delta_M \left(\frac{3M}{M-1} \right)^2. \quad (9.9)$$

All four bounds are $O(M^{-1})$. Summing Poisson pattern coordinates by their type gives independent Poisson variables of mean λ . Coupling each to Poisson(1) costs at most $\lambda - 1$, by adding an independent Poisson increment. Total variation decreases under the summing map, so

$$d_{\text{TV}}(\mathcal{L}(X_n, Y_n, Z_n), \text{Poisson}(1)^{\otimes 3}) \leq B_0 + B_1 + B_2 + B_3 + \frac{3}{M-1} = O(M^{-1}). \quad (9.10)$$

For a numerical bound, multiply the explicit right side by M . Every summand decreases for $M \geq 12$; for the only less immediate term, use

$$MB_3 \leq 72 \left(\frac{M}{M-1} \right)^2 \frac{M}{M-5} \frac{M-4}{M-7}.$$

At $M = 12$ their sum is less than 354, hence less than 400. We have proved, for $n \geq 6$,

$$d_{\text{TV}}(\mathcal{L}(X_n, Y_n, Z_n), \text{Poisson}(1)^{\otimes 3}) \leq \min\{1, 200/n\}. \quad (9.11)$$

The generous constant is not intended as a sharp small- n approximation.

[write] **Checked, 5 October 2026.** In exact rational arithmetic, $M(B_0 + B_1 + B_2 + B_3 + \frac{3}{M-1})$ with the right-hand sides of (9.6)–(9.9) is 353.96 at $M = 12$ (rounded), and 299.25, 236.50, 192.12 at $M = 14, 20, 40$.

Prime cores and a logarithmic cutoff

The published equivalence between simultaneous vanishing of the local counts and absence of the endpoint decomposition sizes, together with (2.4), implies

$$|\mathbb{P}(D_n \text{ indecomposable}) - \mathbb{P}(X_n = Y_n = Z_n = 0)| = O(n^{-1}).$$

Using (9.10) and the symmetry bound, the argument of (6.1) therefore improves to

$$p_n^a = e^{-3} h_n (1 + O(n^{-1})). \quad (9.12)$$

Its implicit constant is independent of its integer argument, so it is uniform for arguments in $[n - \lceil \log n \rceil, n]$.

Put $K = \lceil \log n \rceil$, with natural logarithms. Write $(Y)_K = Y(Y-1) \cdots (Y-K+1)$ for the falling factorial. Any compatible ordered list of K distinct Y indicators forces K distinct matching edges. Thus

$$\begin{aligned} \mathbb{E}(Y_n)_K &\leq (2n)^K \frac{m_{n-K}}{m_n} \\ &= \prod_{j=0}^{K-1} \frac{2n}{2n-2j-1} \leq \exp\{O(K^2/n)\}. \end{aligned} \quad (9.13)$$

Since $(Y_n)_K \geq K!$ on $Y_n \geq K$,

$$\mathbb{P}(Y_n > K) \leq \frac{\exp\{O(K^2/n)\}}{K!} = o(n^{-a}) \quad (9.14)$$

for every fixed $a > 0$. This is a direct growing-order estimate, not an unsupported use of fixed-moment convergence.

The deletion bound (3.7) also extends to $r \leq K$. Uniformly over these r ,

$$\binom{2n}{2r} m_r \leq \exp\{O((\log n)^2)\}, \quad m_{n-r}/m_n \leq 1.$$

By (3.3), $S_N/m_N \leq \exp\{-\frac{1}{2}N \log N + O(N)\}$. Apply it at $N = n - r$ and sum over $r \leq K$ to obtain

$$\mathbb{P}(\text{some deletion of at most } K \text{ chords is symmetric}) \leq \exp\{-\frac{1}{2}n \log n + O(n + (\log n)^2)\}. \quad (9.15)$$

This remains smaller than every fixed inverse power of n . The bounds for long leaves and repeated parents in Section 4 already have order $O(n^{-1})$.

The quantitative sandwich

Use the definition of good connected matchings with the new $K = \lceil \log n \rceil$. Equations (9.14) and (9.15), and the leaf regularity estimates, show that *connected nongood matchings* have unconditional probability $O(n^{-1})$. The leaf-movement proof applies unchanged, since $n - K \rightarrow \infty$. Consequently

$$\begin{aligned} \frac{c_n}{h_n} &\leq \mathbb{E}[\mathbf{1}_{X_n=0} 2^{-Y_n}] + O(n^{-1}) + \frac{4nS_n}{m_n} \\ &= e^{-3/2} + O(n^{-1}). \end{aligned} \quad (9.16)$$

The last equality follows from (9.10), because the integrand lies in $[0, 1]$.

For the lower bound, canonical recovery still gives (6.3), now for this growing K . The exact ratio for $0 \leq k \leq K$ is

$$\frac{3^k \binom{n-k}{k} h_{n-k}}{h_n} = \frac{(3/2)^k}{k!} \frac{n}{n-k} \prod_{j=0}^{k-1} \frac{1 - (k+j)/n}{1 - (j+1/2)/n}. \quad (9.17)$$

For large enough n every factor is positive and its logarithm is bounded using $|\log(1-z)| \leq 2|z|$ for $|z| \leq 1/2$. The factors after $(3/2)^k/k!$ are therefore $\exp\{O((k+1)^2/n)\} = 1 + O((k+1)^2/n)$ uniformly for $k \leq K$. Combining with (9.12) yields

$$\frac{3^k \binom{n-k}{k} p_{n-k}^a}{h_n} = e^{-3} \frac{(3/2)^k}{k!} \left(1 + O\left(\frac{(k+1)^2}{n}\right) \right). \quad (9.18)$$

Importantly, sum this error against the factorial weights:

$$\sum_{k \geq 0} \frac{(3/2)^k}{k!} (k+1)^2 < \infty, \quad \sum_{k > K} \frac{(3/2)^k}{k!} = o(n^{-a})$$

for every fixed $a > 0$. Hence

$$\frac{c_n}{h_n} \geq e^{-3/2} - O(n^{-1}). \quad (9.19)$$

The upper and lower inequalities give the connected estimate in (9.1). Equation (7.2) gives the same estimate for g_n . Finally,

$$\frac{g_{n-1}}{g_n} = \frac{n}{(n-1)(2n-1)}(1 + O(n^{-1})) = \frac{1}{2n} + O(n^{-2}),$$

and dividing (1.4) by g_n proves the strengthened connectivity correction.

Quantitative saturation and automorphisms

For $k = \ell + t + f \leq K$, the exact family count (8.1), the same product bound and the now quantitative denominator give, uniformly over these triples,

$$\mathbb{P}(G \in \mathcal{D}_{n;\ell,t,f}) = e^{-3/2} \frac{2^{-k}}{\ell!t!f!} \left(1 + O\left(\frac{(k+1)^2}{n}\right) \right). \quad (9.20)$$

This holds for either uniform connected or uniform all graphs. Sum the absolute errors over $k \leq K$. The limiting deficit is $\text{Poisson}(3/2)$ and has a finite second moment, so the sum is $O(n^{-1})$. Its tail beyond K is superpolynomially small. Thus the probability of the complement of the truncated family, and in particular the complement of the full decorated family, is $O(n^{-1})$.

Give atypical graphs any defect-count convention, or a cemetery value. Their total probability is $O(n^{-1})$; the tail and summed errors prove the stated total-variation bounds for defect counts and core deficit. On the family, $\text{Aut}(G) \cong (C_2)^{t+f}$ exactly, so bounded indicators give

$$\mathbb{P}(G \text{ asymmetric}) = e^{-1} + O(n^{-1}), \quad \mathbb{P}(G \text{ leafless}) = e^{-1/2} + O(n^{-1}),$$

and $\mathbb{P}(G \text{ split-prime}) = e^{-3/2} + O(n^{-1})$. The group-order distribution is within $O(n^{-1})$ total variation of 2^P , $P \sim \text{Poisson}(1)$. These assertions do not claim convergence of unbounded automorphism moments. The bounded reciprocal automorphism factor similarly gives

$$\frac{g_n^{\text{lab}}}{n!g_n} = e^{-1/2} + O(n^{-1}).$$

Multiply by the quantitative unlabeled equivalent to obtain

$$g_n^{\text{lab}}, c_n^{\text{lab}} = e^{-2} \frac{(2n)!}{2^{n+2n}} (1 + O(n^{-1})), \quad (9.21)$$

where the displayed estimate holds for each sequence separately. This completes the proof of Theorem 9.1.

10 Smooth inversion and ceiling-safe thresholds

10.1 A precise smooth model

For either $a_n = c_n$ or $a_n = g_n$, define the positive real model

$$H(t) = \frac{e^{-3/2} 2^t \Gamma(t + 1/2)}{4\sqrt{\pi} t}, \quad t > 0. \quad (10.1)$$

Then $H(n) = e^{-3/2}h_n$ and $a_n/H(n) \rightarrow 1$. For large t , Stirling's expansion gives

$$\begin{aligned}\log H(t) &= t(\log(2t) - 1) - \log t - B - \frac{1}{24t} + O(t^{-3}), \\ B &= \frac{3}{2}(1 + \log 2).\end{aligned}\tag{10.2}$$

The derivative is $\log(2t) - 1/t + O(t^{-2})$, so H is strictly increasing on a sufficiently far real tail. Let ϕ be that eventual real inverse.

For x large, write

$$L = \log x, \quad u = \frac{L}{W_0(2L/e)}, \quad A = \log(2u),\tag{10.3}$$

where W_0 is the positive principal Lambert function. The exact identity $W_0(2L/e)e^{W_0(2L/e)} = 2L/e$ implies

$$u(\log(2u) - 1) = L.\tag{10.4}$$

Proposition 10.1 (Smooth inverse). *As $x \rightarrow \infty$,*

$$\phi(x) = u + 1 + \frac{3 + \log 2}{2 \log(2u)} + O\left(\frac{1}{u \log u}\right).\tag{10.5}$$

Proof. Put $F(t) = t(\log(2t) - 1)$, so $F'(u) = A$ and $F''(u) = 1/u$. By (10.2) and (10.4),

$$\log H(u) = L - \log u - B + O(u^{-1}).$$

The derivative of $\log H$ is comparable to A . Evaluating at u and $u + 2$ shows that the root $\phi(x) = u + d$ has bounded displacement, eventually $0 < d < 2$. Uniformly for bounded d , Taylor expansion gives

$$0 = \log H(u + d) - L = dA - \log u - B + O(u^{-1}).\tag{10.6}$$

Since $\log u = A - \log 2$,

$$d = \frac{\log u + B}{A} + O((uA)^{-1}) = 1 + \frac{B - \log 2}{A} + O((uA)^{-1}).$$

Finally $B - \log 2 = (3 + \log 2)/2$, proving the formula. The $-1/(24u)$ term in (10.2) affects only the stated remainder, not the displayed coefficient. $\square$

The shift $+1$ is forced by the factor $1/n$ in the forward model. It is not an optional change of the OEIS offset.

[write] **Instances of the transseries apparatus; refined in Part II, 5 October 2026.** The equation (10.4) is the factorial core `p0:prop:factorial-core` of the transseries volume [7] with $\kappa = 1$ and $d = \log 2 - 1$, and u is its solution; after taking logarithms it is the Lambert core `p0:thm:lambert-core` with $a = b = 1$ (front matter, “The OEIS entries, and the inverses”). The displacement $1 + \frac{3 + \log 2}{2 \log(2u)}$ is derived here from the Gamma model, not from the volume. Part II's Theorem 23.1 carries the expansion one order further: with Part II's $D = \log(2u)$ (this Part's A) and $B = \frac{3 + \log 2}{2}$ (this Part's $B - \log 2$), the centre (10.12) is $u + 1 + B/D$, and Part II adds $(K_a - B^2/(2D^2))/(uD)$.

10.2 The least integer index

Define

$$N_a(x) = \min\{n \geq 1 : a_n \geq x\}. \quad (10.7)$$

The leading equivalent implies

$$\frac{a_{n+1}}{a_n} \sim \frac{H(n+1)}{H(n)} = (2n+1) \frac{n}{n+1} \sim 2n, \quad (10.8)$$

so a_n is eventually strictly increasing and unbounded. For x exceeding the maximum on a finite initial segment, $N_a(x)$ lies in that monotone tail.

Proposition 10.2 (Fixed-error ceiling bracket). *For each fixed $\eta > 0$ and all sufficiently large x ,*

$$\lceil \phi(xe^{-\eta}) \rceil \leq N_a(x) \leq \lceil \phi(xe^\eta) \rceil. \quad (10.9)$$

Proof. On a sufficiently far integer tail, $e^{-\eta}H(n) \leq a_n \leq e^\eta H(n)$. At $n = N_a(x)$, $a_n \geq x$ implies $H(n) \geq xe^{-\eta}$, giving the lower bound. At $m = \lceil \phi(xe^\eta) \rceil$, monotonicity of H gives $H(m) \geq xe^\eta$ and hence $a_m \geq x$. This proves the upper bound, once both indices are in the stated tail. $\square$

A shrinking bracket can be made explicit as an *existence* statement without assuming an unproved convergence rate. Let

$$r_n = \left| \log \frac{a_n}{H(n)} \right|, \quad \eta(x) = \sup_{n \geq \lfloor u/2 \rfloor} r_n. \quad (10.10)$$

For large enough x this tail supremum is finite. Theorem 9.1 gives $r_n = O(n^{-1})$ and therefore $\eta(x) = O(u^{-1})$. It is defined from the actual sequence; its implicit constant and onset are not supplied as computable finite-input bounds.

To justify applying this moving envelope, first apply Proposition 10.2 with $\eta = 1$. Proposition 10.1 and the derivative estimate show $\phi(xe^{\pm 1}) = u + O(1)$. Thus $N_a(x)$ is eventually larger than $u/2$, and both model endpoints formed with $0 \leq \eta(x) \leq 1$ also lie above $u/2$. The proof of Proposition 10.2 now applies using the uniform bound (10.10), without a circular choice of its starting index.

By the mean value theorem applied to $\log H$ near u ,

$$\phi(xe^{\pm \eta(x)}) = \phi(x) + O\left(\frac{\eta(x)}{\log(2u)}\right). \quad (10.11)$$

Set

$$s(x) = u + 1 + \frac{3 + \log 2}{2 \log(2u)}. \quad (10.12)$$

There exists a nonnegative function $\varepsilon(x) = O(u^{-1})$ such that, for all sufficiently large x ,

$$\left\lceil s(x) - \frac{\varepsilon(x)}{\log(2u)} \right\rceil \leq N_a(x) \leq \left\lceil s(x) + \frac{\varepsilon(x)}{\log(2u)} \right\rceil. \quad (10.13)$$

For example, sufficiently large fixed constants in $\varepsilon(x) = C_1 \eta(x) + C_2/u$ absorb both (10.11) and the model remainder.

Equivalently, there are constants $C_* > 0$ and $x_* > 0$ such that for $x \geq x_*$,

$$\left\lceil s(x) - \frac{C_*}{u \log(2u)} \right\rceil \leq N_a(x) \leq \left\lceil s(x) + \frac{C_*}{u \log(2u)} \right\rceil. \quad (10.14)$$

The width uses both the quantitative forward remainder and the $O(1/(u \log u))$ error of the displayed smooth-model expansion; no smaller truncation error is silently assumed. Eventually the bracket contains at most two adjacent integers. If its interval contains no integer, the two ceilings agree. Near an integer it may retain two possibilities; the leading asymptotic alone cannot resolve them uniformly. In particular, (10.13) is not a license for unconditional nearest-integer rounding, and the report gives no effective onset or numerical bound for $\varepsilon(x)$.

[write] **Refined in Part II, 5 October 2026.** Part II's (23.2) narrows the half-width $C_*/(u \log(2u))$ of (10.14) to $C_a/(u^2 \log(2u))$, with the same qualifications: noneffective constants and onset, and no unconditional rounding. These brackets follow the separation pattern of the transseries volume's `p0:thm:staircase` (2) but are not instances of it, since H does not interpolate a_n (front matter).

11 Exact finite checks and what they establish

11.1 Source counts and scale

Table 1 records the inspected exact initial values. The companion reconciles these with the unlabeled component product

$$\sum_{n \geq 0} g_n z^n = \prod_{j \geq 1} (1 - z^j)^{-c_j}, \quad (11.1)$$

interpreted as a formal power series. It follows because a graph is a multiset of connected components, uniquely up to order. Finite truncation gives exact coefficient tests; convergence of the displayed ordinary generating series is not assumed.

Table 1: Exact inspected graph counts by order. The order-thirteen all-graph value is from OEIS; no connected value at that order is claimed from the inspected table.

n	c_n	g_n
1	1	1
2	1	2
3	2	4
4	6	11
5	21	34
6	110	154
7	789	978
8	8336	9497
9	117283	127954
10	2026331	2165291
11	40302425	42609994
12	892278076	937233306
13	—	22576188846

Inverting the logarithmic derivative of (11.1) using the inspected all-graph value at order thirteen gives $c_{13} = 21593488017$. This is a derived value, checked by the companion, rather than an entry in the inspected connected source table.

For orientation, the connected ratios c_n/h_n at $n = 8, 9, 10, 11, 12$ are approximately 0.131598, 0.122526, 0.123797, 0.128974, and 0.135436. The all-graph ratio at $n = 13$ is approximately 0.148493.

The limiting constant is $e^{-3/2} \approx 0.223130$. These small values have not converged rapidly enough to estimate the constant reliably. They neither prove nor contradict the theorem.

[write] **Reproduced, 5 October 2026.** Euler inversion of (11.1) with the terms of Table 1 gives $c_{13} = 21,593,488,017$ (also reproduced at intake). The value is not in A156808 (revision #9), and nothing was submitted to the OEIS. With it, $c_{13}/h_{13} \approx 0.142029$. Against Part II’s first corrections, $c_n/(e^{-3/2}h_n(1 - \frac{15}{4n})) \approx 0.8829, 0.8946$ and $g_n/(e^{-3/2}h_n(1 - \frac{13}{4n})) \approx 0.8744, 0.8873$ at $n = 12, 13$: like the ratios above, these small orders are far from the limit and neither confirm nor contradict the expansions. The ratios $c_n/g_n \approx 0.952034, 0.956472$ at $n = 12, 13$ exceed $1 - \frac{1}{2n} - \frac{1}{n^2}$ (Part II, (14.4)) by $6.45 \cdot 10^{-4}$ and $8.50 \cdot 10^{-4}$; n^3 times the excess is 1.11 and 1.87, so at these two orders the excess is of the size of an $O(n^{-3})$ remainder but has not settled to a constant.

11.2 The reproducible companion

The newly written standard-library Python companion verifies finite instances of the fixed-endpoint leaf completion formula (4.1), enumerates matching and graph isomorphism classes at stated small orders, and checks explicit independent leaf-move constructions. In each eligible example it checks the graph, every dihedral orbit size, and disjointness of the 2^k orbits. It distinguishes geometric symmetries from graph automorphisms rather than conflating the two counts.

Further exact checks reconcile the published graph counts and the component product, and verify the inverse coefficient algebra. The machine-readable evidence file records all finite ranges and examples. A separate validator recomputes the evidence and rejects semantic alterations; tests run both normally and with Python optimization enabled, so no mathematical checks depend on removable `assert` statements.

The finite tests check implementation, normalization and examples. They cannot establish a limiting Poisson law, rule out every large- n exceptional representation, prove the recovery lemma for all cores, or establish the quantitative asymptotic rates. Those tasks are handled by the mathematical arguments and cited structural inputs. Small forcing-coupling witnesses likewise do not replace the all-size argument of Section 9.

[write] **The companion here, 5 October 2026.** Shipped as `code/150-leading-companion-*.py` and `data/150-leading-companion-*` (the README maps the delivered names). At intake its evidence was regenerated byte for byte and validated on a copy; its unit tests and the release tests need POSIX file calls and were not run on Windows (README, “Rerunning the checks”).

12 Limitations and further work

The quantified sandwich proves a relative $O(n^{-1})$ error but still does not determine the individual first correction coefficient. Several extensions would require additional work.

1. A first individual coefficient requires analysis of the order- $1/n$ exceptional configurations, including long leaves, shared parents, overlapping local decorations and nonlocal splits, and their actual representation fibers. The connectivity correction alone does not identify that coefficient.
2. An all-orders expansion cannot be inferred by freely expanding the decorated-core lower bound. Nongeneric cores and other decompositions must be controlled at the requested order; the current $O(n^{-1})$ exceptional mass cannot simply be dropped.

3. Effective graph-count constants and onsets would make (10.14) a numerical threshold certificate. The explicit $200/n$ bound applies only to the local matching patterns, not the complete graph-count argument.
4. The final 2026 journal text should be reconciled with the 2024 sources before any stronger priority statement. The optional labeled consequence deserves its own source and sequence audit.

The central model-specific conclusions established here are the reciprocal-fiber upper bound, the canonically recoverable lower construction, their exact matching normalization, and the consequences forced by saturation.

[write] **Status of these items, 5 October 2026.** Item 1 is answered by Part II’s Theorems 14.1 and 14.2, by another route than the one sketched here: instead of analysing the exceptional configurations listed, Part II expands the local pattern counts uniformly (Theorem 16.1), controls the nonlocal decompositions by its three-chord interval lemma, and transfers the prime correction to graph classes through canonical split trees. Item 2 is re-scoped: Part II’s conditional transfer (Subsection 21.3) controls the nongeneric cores and the graph splits at every fixed order and reduces the graph expansion to the expansion of p_n^a/h_n , which is known only through its first correction; the next prime coefficient β is open (Section 25, item 1). Items 3 and 4 remain open (items 3, 4 and 5).

13 Dependency map and reproducible release

[write] Report 150’s Appendix A, printed as a numbered section of Part I so that Part II’s sections follow it.

The logical dependencies are as follows.

1. The published matching decomposition and local Poisson lemmas give prime probability e^{-3} ; prime uniqueness and the geometric automorphism action connect this to asymmetric prime graph counts.
2. The direct rotation and two-reflection estimates control all geometric symmetries, including symmetries induced after bounded deletions.
3. Long-leaf and shared-parent bounds plus independent leaf moves bound connected reciprocal fibers from above.
4. Canonical recovery of the three decorations makes the lower count injective, and its normalized exponential sum matches the upper constant.
5. The component convolution transfers the equivalent to all graphs and isolates the connected/all correction. Saturation gives the core and automorphism laws. A direct local-pattern forcing coupling, an elementary Poisson approximation bound and a logarithmic cutoff quantify these rates. Monotone smooth inversion gives the threshold bracket.

The source archive contains the report PDF, editable LaTeX, new companion code and evidence, a source-provenance record, checksums, deterministic build scripts, and release tests. It excludes downloaded papers, private working notes, prior reports, and build intermediates. The README gives the exact commands, required toolchain, finite computation ranges and output-safety rules. PDF byte reproducibility is qualified to the tested pdfTeX toolchain; the mathematical contents are

not dependent on that byte identity. The checksum manifest establishes package consistency, not authenticity against replacement of both a payload and its manifest.

[write] **The release here, 5 October 2026.** The PDF and the checksum manifest **SHA256SUMS** are not shipped (repository policy; the manifest verified 17/17 at intake); the editable source is this Part; the companion, provenance record and release scripts are shipped under prefixed names (README).

Part II

First corrections for unlabeled circle graphs

Bundle Report 152, archive `Circle_Graph_First_Corrections_and_Refined_Inverses_Source.zip`, delivered as `Report152.tex` (1200 lines, 19 pages). Delivered title block “First corrections for unlabeled circle graphs / Connected and all graphs in OEIS A156808 and A156809 / Report 152 / 3 October 2026”. Printed as delivered, with **[write]** notes. Delivered Section k is Section $k + 13$ here, and its statement and equation numbers $k.j$ are $(k + 13).j$. Section 25, which closes the report, is added in the write.

Reading conventions of Part II	Elsewhere in the report
c_r the coefficients of $H_M(u)$ (Section 16)	$c_n = A156808(n)$ everywhere else
$q_r = \prod_{j < r} (M - 2j - 1)^{-1}$	Part I: ρ_r ; Part I's q_s is $m_s m_{n-s} / m_n$
x, y, z complex weights; y also the threshold input	Part I: x the threshold input
$\alpha = -5$, β the prime coefficients; $\gamma = \alpha + \frac{5}{4}$,	Part I: α, β pattern indices; η bracket errors
$\eta = \gamma + \frac{1}{2}$	
$B = \frac{3+\log 2}{2}$ and $D = \log(2u)$ in the inverse	Part I: $B = \frac{3}{2}(1 + \log 2)$, $A = \log(2u)$
$b_j, b_j(t)$ rooted counts; r_s rooted classes; $\rho = 2^{-t}$,	Part I: ρ_r a pattern probability
$\sigma = 6^{-t}$	
$C_n^{\text{lab}}, G_n^{\text{lab}}$	Part I: $c_n^{\text{lab}}, g_n^{\text{lab}}$

Abstract of this Part (as delivered). Let c_n and g_n count connected and arbitrary unlabeled circle graphs on n vertices, and put $h_n = (2n - 1)!! / (4n)$. We prove

$$c_n = e^{-3/2} h_n \left(1 - \frac{15}{4n} + O(n^{-2}) \right), \quad g_n = e^{-3/2} h_n \left(1 - \frac{13}{4n} + O(n^{-2}) \right).$$

A uniform complex generating-function expansion for three local matching patterns gives prime probability $e^{-3}(1 - 5/n + O(n^{-2}))$. A direct three-chord interval lemma controls the non-local remainder. Canonical graph split trees then transfer the prime coefficient to graph-isomorphism classes, with explicit rooted-branch counts and globally controlled tails. We also obtain $c_n/g_n = 1 - 1/(2n) - 1/n^2 + O(n^{-3})$ and a refined Lambert- W inverse with ceiling uncertainty $O(1/(u^2 \log u))$. An every-fixed-order transfer theorem is conditional on the corresponding prime expansion; no unconditional higher prime coefficient is claimed. Exact finite diagnostics and deterministic PDF and source-archive builders accompany the proof.

Source and scope. The primary circle-graph source basis is arXiv:2402.06394v1 (2024), a complete author-hosted 2024 copy, and the structural references below. The 2026 journal version was identified bibliographically; its final text was not inspected. The first corrections established here were not located in the bounded source review. This is not an exhaustive priority claim. The remainders and inverse onsets are asymptotic existence bounds, not effective finite-input certificates. No external publication, author contact, or public repository change is part of this report.

[write] Reading Part II against Part I, 5 October 2026. “The earlier Report 150” (Subsection 15.2) is Part I. Part II's $c_n, g_n, m_n, h_n, \mathcal{M}_n, S_n, R_n(G), p_n^a, X, Y, Z$ and $M = 2n$ are Part I's. Part II cites Part I only in that priority paragraph: it re-proves Part I's symmetry and coarse bounds (Subsection 15.1) and proves its local expansion without the qualitative Poisson limit Part I starts from. Its Theorem 14.1 supersedes the count statements of Part I's Theorem 9.1, and its inverse refines Part I's Section 10.

14 Results and conventions

A circle graph is the intersection graph of chords with distinct endpoints on a circle. All graphs are finite, simple and undirected; “unlabeled” means one graph-isomorphism class, counted once. Write

$$c_n = \text{A156808}(n), \quad g_n = \text{A156809}(n), \quad m_n = (2n-1)!! = \frac{(2n)!}{2^n n!}, \quad h_n = \frac{m_n}{4n}.$$

The two OEIS offsets are one. We use $m_0 = g_0 = 1$ for the empty matching and graph. An indexed matching has its $2n$ endpoints in fixed cyclic order, but no independent labels on its chords. The dihedral group on the endpoints has order $4n$. Geometric symmetry of a matching and abstract symmetry of its graph are different notions; we identify them only for the prime class where justified.

A graph split is a partition $A \sqcup B$ with $|A|, |B| \geq 2$ whose crossing edges form $A' \times B'$ for subsets of the two sides. A nondegenerate split-prime graph has no such split; cliques and stars are treated as degenerate labels in split trees. Let p_n^a count abstractly asymmetric prime circle graphs. All assertions about primes concern sufficiently large orders, so exceptional small conventions play no role.

Theorem 14.1 (Individual first corrections). *As $n \rightarrow \infty$,*

$$p_n^a = e^{-3} h_n \left(1 - \frac{5}{n} + O(n^{-2}) \right), \quad (14.1)$$

$$c_n = e^{-3/2} h_n \left(1 - \frac{15}{4n} + O(n^{-2}) \right), \quad (14.2)$$

$$g_n = e^{-3/2} h_n \left(1 - \frac{13}{4n} + O(n^{-2}) \right). \quad (14.3)$$

Equivalently, with $Q_n = e^{-3/2} (2n/e)^n / (2\sqrt{2}n)$,

$$c_n = Q_n \left(1 - \frac{91}{24n} + O(n^{-2}) \right), \quad g_n = Q_n \left(1 - \frac{79}{24n} + O(n^{-2}) \right).$$

The latter coefficients include the $-1/(24n)$ correction in m_n ; they are not the coefficients in the h_n normalization.

[write] **Relation to Part I, 5 October 2026.** With $h_n = Q_n(1 - \frac{1}{24n} + O(n^{-2}))$, the two normalizations differ by $-\frac{1}{24}$ in the first coefficient. Both expansions imply Part I’s Theorems 1.1 and 9.1 for the counts.

Theorem 14.2 (Connectivity and conditional higher transfer). *Unconditionally,*

$$\frac{c_n}{g_n} = 1 - \frac{1}{2n} - \frac{1}{n^2} + O(n^{-3}). \quad (14.4)$$

For each fixed $q \geq 0$, an expansion of p_n^a/h_n through n^{-q} with remainder $O(n^{-q-1})$ transfers to such expansions of both c_n/h_n and g_n/h_n . This is a conditional transfer theorem, not an unconditional full expansion. In particular, if the additional hypothesis

$$p_n^a = e^{-3} h_n \left(1 - \frac{5}{n} + \frac{\beta}{n^2} + O(n^{-3}) \right)$$

holds for some β , then the second relative coefficients are $\beta - 277/32$ for c_n and $\beta - 297/32$ for g_n . No value of β is proved or conjectured here.

The main two proof steps are different. The local-pattern analysis counts indexed matchings. The graph transfer counts unlabeled isomorphism classes via their canonical split trees. In particular, we never assume that a graph split of a specified size is an interval split of that same size in every chord representation.

15 Structural source inputs and a coarse bound

Bassino, Bouvel, Féray, Gerin and Pierrot [1, Definition 5.3, Proposition 5.5] define a matching decomposition by four successive circular intervals, with every chord inside one interval or joining opposite intervals. Empty intervals are allowed. In their anchored k -decomposition, C_1 contains a fixed endpoint and the k chords lie in $C_2 \cup C_4$. Their structural equivalence says that an indecomposable matching has a split-prime intersection graph and conversely. We use their Lemma 5.9 to identify zero local defects with the absence of the two endpoint decomposition sizes, and Lemma 5.6's exact count

$$d_n^k = (n - k)m_{k+1}m_{n-k+1} \quad (15.1)$$

for pairs consisting of a matching and one chosen k -decomposition. It upper-bounds the number of distinct matchings admitting such a decomposition; their Lemma 5.7 uses it to bound the summed probability. Their Proposition 5.12, attributed there to Gabor, Supowit and Hsu, gives uniqueness of a prime representation up to rotation and reflection for order at least five.

Klavík and Zeman [4, Section 3, Lemmas 7–8 and their proofs] give closure of circle graphs under splitting and inverse split composition. They identify graph automorphisms with canonical split-tree automorphisms and describe prime graph automorphisms as rotations or reflections of the unique circle representation. The action on chords is faithful in prime order at least five. Indeed, a rotation fixing every chord must be a half-turn with all chords diameters, yielding a clique. A reflection fixing every chord gives noncrossing chords perpendicular to its axis, with at most one additional axis chord; its graph is a star with possible isolates. Neither graph is prime at this order. Thus prime abstract asymmetry is equivalent to geometric asymmetry.

Gioan and Paul [3, Section 2.2, Theorem 2.9] give the unique reduced graph-labelled split tree of a connected graph. Its internal nodes have degree at least three, with prime, clique or star labels. Adjacent clique labels and the reducible center-to-leaf star joins are forbidden. Their Lemma 2.8 says each internal tree edge induces a graph split. Corollary 2.10 says every graph split is induced by an edge after at most one split of a degenerate node. Corollary 2.6 identifies labels as induced subgraphs; Lemma 2.3 supplies connectedness of the accessibility graph when all labels are connected. These precise source inputs are what the graph-level proof needs.

15.1 Symmetries are negligible to every fixed order

Let S_n be the number of indexed matchings fixed by a nonidentity rotation or reflection. A rotation of order $d \geq 2$ has $k = 2n/d$ endpoint cycles. Its invariant matchings number

$$k! [z^k] \exp \left(\mathbf{1}_{2|d} z + \frac{d}{2} z^2 \right).$$

Two different cycles are paired with one of d offsets; an internally matched cycle must have even order and pair opposite points. Evaluating the nonnegative series at $z = \sqrt{k/d}$ and using $k! \leq k^k$ bounds this count by

$$(kd)^{k/2} \exp(k/2 + \sqrt{k/d}) \leq (2en)^{n/2} \exp(\sqrt{n}).$$

A gap-axis reflection has the same count as $d = 2, k = n$. An endpoint-axis reflection forces its two fixed endpoints to match together, leaving the case $d = 2, k = n - 1$. Therefore

$$S_n \leq 4n(2en)^{n/2} e^{\sqrt{n}} = o(m_n n^{-A}) \quad \text{for every fixed } A > 0. \quad (15.2)$$

The last claim follows from $\log m_n = n \log(2n) - n + O(1)$.

If $R_n(G)$ is the number of indexed representations of G , then grouping matchings by their graph gives exactly

$$g_n = \sum_{D \in \mathcal{M}_n} \frac{1}{R_n(G_D)}.$$

For a geometrically asymmetric D , its orbit has $4n$ members, so $R_n(G_D) \geq 4n$; on the other matchings use $1/R_n \leq 1$. Consequently

$$c_n \leq g_n \leq h_n + S_n = O(h_n), \quad g_s \leq C \frac{m_s}{s} \quad (s \geq 1) \quad (15.3)$$

for a constant enlarged over finitely many small orders. This short argument makes the new transfer proof independent of any unproved strengthening of earlier leading estimates.

[write] **Restated from Part I, 5 October 2026.** (15.2) is Part I's Lemma 3.1 with (3.3), by the same argument; the fibre identity and (15.3) are Part I's (3.4), (3.5) and (3.6), with $c_n \leq g_n$ added. The method is that of [1, Proposition 5.14]; see Remark 3.3. The faithfulness argument of the preceding paragraphs is Part I's, before (2.2).

15.2 Bounded source priority

The complete 2024 arXiv v1 and the complete author-hosted 2024 copy were inspected in the preceding source review; the relevant text agrees. The journal metadata is *Electronic Journal of Probability* **31** (2026), article 110, [doi:10.1214/26-EJP1570](https://doi.org/10.1214/26-EJP1570). Its final text was not inspected. The earlier Report 150 established the leading equivalent and relative $O(1/n)$ bounds; this report identifies the individual first coefficients with stronger errors. Neither that comparison nor the bounded literature search certifies universal novelty. The source archive records the reviewed versions, source hashes, and proof dependencies.

[write] **Sources, 5 October 2026.** As in Part I (notes in Subsections 2.2 and 2.3): arXiv:2402.06394v1 is the only arXiv version as of 5 October 2026 and was read at the write; the journal text was not read. The write also read Gioan and Paul's statements cited in this section in arXiv:0810.1823v2, where they carry the same numbers; the published version was not compared.

16 Local atoms and the exact polymer representation

Put $M = 2n$, and work on the cycle with vertex set $\mathbb{Z}/M\mathbb{Z}$. An X atom requires a chord between consecutive endpoints. A Y atom requires a chord at cyclic distance two. A Z atom requires one of the two matchings between two disjoint consecutive endpoint pairs. Count each such specified two-chord configuration once. For $M \geq 6$ these give M atoms of each singleton type and $M(M-3)$ atoms of type Z . Let X, Y, Z be the counts of atoms present in a uniform matching. The letters x, y, z below are complex weights, not counting variables for vertices.

Theorem 16.1 (Uniform local expansion). *Uniformly for (x, y, z) in every fixed compact subset of $\mathbb{C}^3$,*

$$\mathbb{E}[(1+x)^X (1+y)^Y (1+z)^Z] = e^{x+y+z} \left(1 + \frac{F(x, y, z)}{M} + O(M^{-2}) \right), \quad (16.1)$$

where

$$F = x + y + z - \frac{1}{2}x^2 - \frac{1}{2}y^2 - z^2 - 2xy - xz - yz + y^2z. \quad (16.2)$$

In particular,

$$\mathbb{P}(X = Y = Z = 0) = e^{-3} \left(1 - \frac{10}{M} + O(M^{-2}) \right). \quad (16.3)$$

[write] **Monte Carlo, uncertified, 5 October 2026.** At intake, $\mathbb{P}(X = Y = Z = 0)$ was estimated on 200,000 uniform matchings for each of $M = 100, 200, 400$ (NumPy, seed 20261005; intake record, not shipped). The rescaled estimates of the coefficient of $1/M$, namely $M(\hat{P}/e^{-3} - 1)$, are -9.88 ± 0.93 , -9.83 ± 1.9 and -4.4 ± 3.9 , with one binomial standard error. The first two agree with -10 and exclude 0 by about 10 and 5 standard errors; at $M = 400$ the estimate lies within 1.5 standard errors of both -10 and 0. This is evidence for a proved statement, and it says nothing about the second-order term.

16.1 Exact partition polynomial

An atom set A is compatible if its required chords form a matching. Write $r(A)$ for their number of distinct chords and $t(A)$ for the product of the atom weights. Define

$$H_M(u) = \sum_{A \text{ compatible}} t(A) u^{r(A)} = \sum_{r=0}^n c_r u^r. \quad (16.4)$$

Expansion of the product of atom indicators is exact and gives

$$\mathbb{E}[(1+x)^X (1+y)^Y (1+z)^Z] = \sum_{r=0}^n c_r q_r, \quad q_r = \prod_{j=0}^{r-1} (M - 2j - 1)^{-1}. \quad (16.5)$$

Here q_r is the probability of r specified disjoint chords.

[write] In this section and the next, c_r is a coefficient of $H_M(u)$, not A156808(r).

Connect two atoms when their endpoint supports meet. Inside a compatible set this means they share a whole chord. A polymer is a nonempty compatible atom set connected under this relation. Its support is denoted S_P , size $r(P)$ counts its distinct chords, and activity is $w_P = t(P)u^{r(P)}$. Compatible atom sets have unique connected-component decompositions into endpoint-disjoint polymers; conversely such a family reconstructs the atom set. Hence H_M is exactly the hard-core polymer partition polynomial, where overlapping supports, including identical polymers, conflict.

16.2 Uniform counting and absolute tails

In the graph of possible chords, join two chords when together they form a Z atom. Each chord has at most four neighbors: choose one cycle neighbor of each endpoint. A connected chord set of size $r \geq 2$ with a specified root chord has at most 4^{2r-2} possibilities. To see this, choose a canonical spanning tree and its depth-first traversal of length $2r - 2$; each step has at most four choices and the visited set determines the chord set. There are at most $2r$ possible Z edges on that set, and at most r possible singleton atoms. Thus at most 2^{3r} atom subsets occur on it. Bounded complex weights can be absorbed into C^r . For a constant depending only on the fixed weight compact set:

- There are at most $M^2 C^r$ polymers of size $r \geq 2$
- At most $M C^r$ of them contain any specified endpoint, by rooting at its unique incident chord
- There are exactly $2M$ size-one polymers

- At most $2MC^r$ polymers of size r contain a singleton atom, by rooting at that singleton chord

For every fixed S and nonnegative integer k , these estimates and geometric summation give, uniformly for $|u| \leq S/M$,

$$\sum_P r(P)^k |w_P| = O(1), \quad \sup_v \sum_{P: v \in S_P} r(P)^k |w_P| = O(M^{-1}). \quad (16.6)$$

The total activity of size at least four is $O(M^{-2})$, with any fixed polynomial size weight. So is that of size at least three with a singleton atom. In addition, dropping endpoint restrictions in the positive-activity polymer sum yields

$$\sum_r |c_r| (S/M)^r \leq \sum_{A \text{ compatible}} |t(A)| (S/M)^{r(A)} \leq \exp \left(\sum_P |t(P)| (S/M)^{r(P)} \right) = O(1). \quad (16.7)$$

The fact that this holds for every fixed S is essential. Fixed factorial moments alone would not justify the later summation over all r or evaluation at the zero-defect weights.

17 The first hard-core correction

For fixed u , put

$$W = \sum_P w_P, \quad B = \sum_{P \sim Q} w_P w_Q,$$

where the latter sum is ordered and includes $P = Q$. Then

$$H_M(u) = e^W (1 - B/2) + O(M^{-2}) \quad (|u| \leq S/M). \quad (17.1)$$

Here is an elementary remainder proof valid for complex activities. Express the partition function as the sum over ordered polymer k -tuples, divided by $k!$, with the indicator of no conflicting index pairs. If N is the number of such pairs in a tuple, then

$$\left| \mathbf{1}_{\{N=0\}} - (1 - N) \right| \leq \binom{N}{2}.$$

The terms 1 and $-N$ sum to e^W and $-e^W B/2$. Two distinct conflict pairs use either four or three indices. With absolute activities the resulting remainder is bounded by

$$e^{W_*} (B_*^2/8 + C_*/2), \quad (17.2)$$

where

$$W_* = \sum_P |w_P|, \quad B_* = \sum_{P \sim Q} |w_P w_Q|, \quad C_* = \sum_P |w_P| \left(\sum_{Q \sim P} |w_Q| \right)^2.$$

By endpoint incidence in (16.6), $\sum_{Q \sim P} |w_Q| = O(r(P)/M)$. Therefore $W_* = O(1)$, $B_* = O(M^{-1})$ and $C_* = O(M^{-2})$. This proves (17.1) without appealing to a formal infinite cluster series or assuming positivity of the original weights.

17.1 All relevant connected polymers

Set $u = s/M$ for bounded complex s . Size-one polymers contribute $s(x + y)$; single- Z polymers contribute $zs^2 - 3zs^2/M$. The other two-chord polymers are:

Atom types	Number	Reason
XZ	M	The partner of $\{i, i + 1\}$ is $\{i - 1, i + 2\}$
YZ	$3M$	Three partners of each $\{i, i + 2\}$
YYZ	M	$\{i, i + 2\}, \{i + 1, i + 3\}$

For the Y chord the three partners are $\{i - 1, i + 1\}, \{i - 1, i + 3\}, \{i + 1, i + 3\}$. Distinct singleton atoms cannot share a chord. This list is therefore exhaustive on two chords. In particular, the two YZ subpolymers in a YYZ configuration are different from its full three-atom polymer; all must be counted. Their combined extra activity is $s^2(xz + 3yz + y^2z)/M$.

The Z -adjacency graph of any compatible matching has maximum degree two. Two compatible neighbors of a chord must use the two different cycle neighbors of each endpoint. A cycle of such adjacencies therefore has endpoint support closed under neighbors of the original cycle, and hence uses all M endpoints. Thus three-chord Z cycles are impossible for $M \geq 8$. The three-chord polymers without singleton atoms are paths of two Z edges. Their number is exactly $M(M - 5)$. Indeed the common chord has distance at least three in both directions, giving $M(M - 5)/2$ choices; its four neighboring endpoints have two compatible pairings. The common chord is unique. This contributes $z^2s^3/M + O(M^{-2})$. Section 16.2 controls every remaining polymer. Hence

$$\begin{aligned} W &= A(s) + W_1(s)/M + O(M^{-2}), \\ A(s) &= s(x + y) + s^2z, \\ W_1(s) &= s^2(-3z + xz + 3yz + y^2z) + s^3z^2. \end{aligned} \tag{17.3}$$

Both the YYZ term and the shared-chord ZZ path contribute at first order. Truncating original-atom clusters at pairs would miss one of them.

17.2 Endpoint exclusions

A polymer is elementary if it contains one original atom. Non-elementary polymers have total absolute activity $O(M^{-1})$, even with a polynomial size weight. Their contribution to B is $O(M^{-2})$ by endpoint incidence. Elementary supports have the following ordered overlap counts, including identical atoms:

type order	XX	YY	XY	XZ	YZ	ZZ
count	$3M$	$3M$	$4M$	$6M^2 + O(M)$	$8M^2 + O(M)$	$12M^3 + O(M^2)$

(17.4)

The first three follow by fixing the first singleton and choosing one of three, three or four intersecting singleton supports. Every endpoint lies in $4(M - 3)$ Z supports. An adjacent pair lies in $2M - 4$ of them, so an X support meets $6M - 20$ Z atoms. A distance-two pair lies in six Z supports, so a Y support meets $8M - 30$ of them.

For the last leading constant, inclusion–exclusion over shared endpoints starts with $M[4(M - 3)]^2 = 16M^3 + O(M^2)$. Subtracting shared adjacent pairs removes $M(2M - 4)^2 = 4M^3 + O(M^2)$. Nonadjacent pairs are in at most a fixed number of Z supports, so their total is $O(M^2)$. Any triple

or quadruple is also in only a fixed number of Z supports, and only $O(M^2)$ such sets appear in any support. Thus the remaining inclusion–exclusion terms are $O(M^2)$, proving the $12M^3$ leading term. Consequently

$$\begin{aligned} B &= B_1(s)/M + O(M^{-2}), \\ B_1(s) &= s^2(3x^2 + 3y^2 + 8xy) + s^3(12xz + 16yz) + 12s^4z^2. \end{aligned} \quad (17.5)$$

Combining (17.1), (17.3) and (17.5),

$$H_M(s/M) = e^{A(s)} \left(1 + \frac{W_1(s) - B_1(s)/2}{M} \right) + O(M^{-2}), \quad (17.6)$$

uniformly on every fixed complex compact set of the weights and s .

18 Exact matching probabilities and the prime correction

18.1 A global remainder through the full matching range

There is an absolute constant C such that, for $0 \leq r \leq M/2$,

$$q_r = M^{-r} \left(1 + \frac{r^2}{M} + R_{r,M} \right), \quad |R_{r,M}| \leq \frac{Cr^4 e^r}{M^2}, \quad R_{0,M} = 0. \quad (18.1)$$

To prove this, put

$$a_{r,M} = \log(q_r M^r) = \sum_{j=0}^{r-1} -\log \left(1 - \frac{2j+1}{M} \right).$$

Convexity and the midpoint-integral inequality give

$$a_{r,M} \leq \frac{M}{2} \int_0^{2r/M} -\log(1-v) dv \leq r,$$

including $r = M/2$ by the integrable endpoint limit. For $r \leq M/4$, Taylor's formula for $-\log(1-v)$ gives

$$0 \leq a_{r,M} - r^2/M \leq Cr^3/M^2, \quad a_{r,M} \leq 2r^2/M.$$

Using $|e^a - 1 - a| \leq a^2 e^a/2$ proves (18.1) in this range. For $r > M/4$, use $e^a \leq e^r$ and $r^4/M^2 \geq 1$ for large M , increasing C for the finite remaining cases. Thus the estimate is global, not merely a fixed- r approximation.

Insert this into (16.5). Since $r^4 e^r \leq C' e^{2r}$, (16.7) with $S = e^2$ makes the total error $O(M^{-2})$. Writing $\mathcal{D} = s d/ds$,

$$\sum_r c_r q_r = H_M(1/M) + \frac{1}{M} \mathcal{D}^2 H_M(s/M) \Big|_{s=1} + O(M^{-2}). \quad (18.2)$$

Uniform complex analyticity of (17.6) near $s = 1$ permits differentiation of its leading approximation by Cauchy's estimate. Therefore

$$\mathcal{D}^2 H_M(s/M) \Big|_{s=1} = e^{x+y+z} ((x+y+2z)^2 + x+y+4z) + O(M^{-1}).$$

The relative coefficient in (18.2) is

$$W_1(1) - B_1(1)/2 + (x + y + 2z)^2 + x + y + 4z = F(x, y, z).$$

This proves Theorem 16.1; at $x = y = z = -1$, $F = -10$. For example, the univariate consequence is

$$\mathbb{E}(1+t)^{X+Y+Z} = e^{3t} \left(1 + \frac{3t - 6t^2 + t^3}{M} + O(M^{-2}) \right).$$

This identity follows from uniform analyticity; fixed cumulants were not used as a substitute for the global bounds.

18.2 A direct three-chord interval lemma

Lemma 18.1. *If all endpoints of three chords occupy the union of two circular intervals containing no endpoints of other chords, then those three chords contain an X , Y or Z atom.*

Proof. Swap intervals so their endpoint counts are $0 + 6$, $1 + 5$, $2 + 4$ or $3 + 3$. An internal chord of within-interval distance one or two already gives X or Y , so suppose none exists.

- In $0 + 6$, positions $2, 1, 0$ must match respectively $5, 4, 3$. Two parallel consecutive chords give Z
- In $1 + 5$, the only two disjoint internal chords of length at least three are $\{0, 3\}$ and $\{1, 4\}$; these give Z
- In $2 + 4$, both endpoints of the shorter interval match into the longer one. Its remaining internal chord must be $\{0, 3\}$; positions $1, 2$ match to the adjacent shorter-interval pair, giving Z
- In $3 + 3$, all chords join the intervals. The two adjacent pairs of the first interval have distinct image pairs; only one pair of the second interval is nonadjacent. At least one image pair is adjacent, giving Z

Extra adjacency across an empty separating interval can only add local atoms, so the proof covers that case too. $\square$

A 3- or $(n - 3)$ -decomposition has one side satisfying the lemma, while source Lemma 5.9 excludes 2 and $n - 2$ when $X = Y = Z = 0$. Thus a zero-defect decomposable matching must have a k -decomposition with $4 \leq k \leq n - 4$. Set $e_n^k = m_{k+1}m_{n-k+1}$; it is symmetric and decreases towards the midpoint because

$$\frac{e_n^{k+1}}{e_n^k} = \frac{2k + 3}{2n - 2k + 1}.$$

The source bound (15.1) and a union bound give

$$\begin{aligned} \sum_{k=4}^{n-4} \frac{d_n^k}{m_n} &\leq n \left(2 \frac{e_n^4}{m_n} + n \frac{e_n^5}{m_n} \right) \\ &= n(O(n^{-3}) + nO(n^{-4})) = O(n^{-2}). \end{aligned}$$

This is a statement about decompositions of the actual matching, with no graph-split-size assumption. Consequently

$$\mathbb{P}(D_n \text{ indecomposable}) = \mathbb{P}(X = Y = Z = 0) + O(n^{-2}) = e^{-3} \left(1 - \frac{5}{n} + O(n^{-2}) \right). \quad (18.3)$$

The difference from the zero-defect probability is nonpositive. Removing the at most S_n symmetric matchings and dividing each remaining prime orbit by exactly $4n$ proves (14.1), by prime uniqueness and (15.2).

[write] **Credit and relation to Part I, 5 October 2026.** The ratio e_n^{k+1}/e_n^k , the monotonicity towards the midpoint and the bound of the middle terms by their extreme values are those of the source's proof of its Lemma 5.7 (with the same notation e_n^k). Part II's sum starts at $k = 4$, because Lemma 18.1 excludes $k = 3$ and $k = n - 3$ for zero-defect matchings, and it isolates the terms $k = 4$ and $k = n - 4$; together these turn the source's $O(n^{-1})$ into $O(n^{-2})$. The count (15.1) is its Lemma 5.6 (both checked in arXiv:2402.06394v1). (18.3) gives Part I's (2.6) its first correction.

19 A graph level finite split tree formula

Let r_s be the number of connected circle-graph isomorphism classes with one distinguished vertex. Rooting means choosing a vertex orbit, so (15.3) gives

$$r_s \leq sc_s \leq Cm_s. \quad (19.1)$$

Put $b_j = r_{j+2}$ and $B_d(z) = \sum_{j=0}^d b_j z^j$.

Theorem 19.1 (Finite canonical transfer). *For every fixed integer $d \geq 1$,*

$$c_n = \sum_{N=\lceil n/(d+1) \rceil}^n p_N^a[z^{n-N}] B_d(z)^N + O(h_n n^{-d}). \quad (19.2)$$

Apart from a superpolynomially negligible symmetric-core class, the sum counts exactly the connected graphs with no split whose two sides both have at least $d + 2$ vertices, for all sufficiently large n .

19.1 Large split errors without choosing a representation

A split of a connected circle graph with sides $s, n - s$ produces rooted factors of sizes $s + 1, n - s + 1$ by adjoining a marker adjacent to the frontier on its side. Both factors are connected. Each is isomorphic to an induced subgraph of the original circle graph: use any vertex in the opposite nonempty frontier for the marker. The two rooted isomorphism classes determine their split composition uniquely, by deleting both roots and joining their neighbor sets completely. Uniqueness of the original split is unnecessary for an upper bound.

Let $E_{n,r}$ count connected circle graphs with a split whose sides both have at least $r \geq 2$ vertices. Then

$$E_{n,r} \leq \sum_{s=r}^{\lfloor n/2 \rfloor} r_{s+1} r_{n-s+1} \leq C^2 \sum_{s=r}^{\lfloor n/2 \rfloor} m_{s+1} m_{n-s+1} = O(h_n n^{-r+2}). \quad (19.3)$$

For the last equality put $a_s = m_{s+1} m_{n-s+1}$. The ratio $a_{s+1}/a_s = (2s + 3)/(2n - 2s + 1)$ is at most one in the needed half range, so the sum is bounded by $a_r + n a_{r+1} = O(a_r) = O(m_{n-r+1})$ for fixed r . Finally $m_{n-r+1}/m_n = O(n^{-r+1})$. In particular, every graph with a split whose sides both have at least four vertices contributes in total $O(h_n/n^2)$. The proof operates on graph factors, not interval representations.

19.2 The canonical large prime center

Fix $r \geq 3$ and suppose no graph split has two sides of size at least r . Give each leaf of the canonical reduced split tree weight one and each internal node weight zero. A weighted centroid exists with every component after its removal containing at most $n/2$ leaves: repeatedly move towards a component of weight larger than $n/2$. This process advances along the finite tree, so stops, and for $n > 2$ stops at an internal node v . Every component at v has at most $r - 1$ leaves. Otherwise its internal attachment edge induces a forbidden split; a component attached by a leaf edge has size one. If v has degree N , then

$$N \geq n/(r - 1). \quad (19.4)$$

For $n \geq 3r - 2$, v cannot be degenerate. Greedily group its ports until their components total between r and $2r - 2$ leaves. This group and its complement each use at least two ports, and each carry at least r leaves. Any partition of a clique or star label into two sets of at least two markers is a split. Splitting that degenerate node would thus give a forbidden graph split, a contradiction.

For $n > r(r - 1)$ the prime center has degree $N > r$. Every other internal node is inside a branch of at most $r - 1$ leaves. Adjoining its attachment marker gives at most r leaves, so such a node has degree at most r , since each of its incident components contains a leaf. The center is therefore the unique node of degree greater than r and is canonically recovered by graph isomorphisms.

Conversely, attach branches of at most $r - 1$ actual vertices to a prime center of degree $N > r$. By source Corollary 2.10, any split comes from a tree edge or one degenerate-node split. The prime center cannot split; every degenerate node is inside one branch. In each case the side away from the center has at most $r - 1$ leaves. Thus no split has two sides of size at least r . This proves the needed characterization, including its converse.

19.3 Rooted branches and isomorphism multiplicities

For each port x of a prime graph H , choose a rooted connected circle graph (S_x, ρ_x) of order $j + 2$ with $0 \leq j \leq d$. Its actual branch vertices are $V(S_x) \setminus \{\rho_x\}$, and its excess over the core port is j . With $A_x = N_{S_x}(\rho_x)$, retain all within-branch edges and put all edges of $A_x \times A_y$ between branches exactly when xy is an edge of H . This is simultaneous split composition.

The identity branch is rooted K_2 . There is no order-one branch, which would leave zero actual vertices after root deletion, and no second distinguished attachment vertex or marker orientation. At the tree level, remove the root leaf from the reduced tree of S_x and connect its former neighbor to the center port. For K_2 simply attach the actual leaf. All old adjacencies remain reduced, and each new internal adjacency has a prime endpoint; therefore no reduction changes the core or the branch classes. Cutting a branch and restoring its root leaf recovers exactly (S_x, ρ_x) .

If H is asymmetric, assignments give the same unmarked graph exactly when each rooted branch type agrees at the same port. Thus the exact contribution of H is $[z^{n-N}]B_d(z)^N$. There is no division by $N!$, no extra root count, and no division by branch automorphism groups: rooted isomorphism types already account for them. Indeed

$$\text{Aut}(G) \cong \prod_{x \in V(H)} \text{Aut}(S_x, \rho_x) \quad \text{when } \text{Aut}(H) = 1.$$

For a symmetric prime center the exact count would require its port-action orbits; the upper bound

$B_d(1)^N$ suffices here. Using (15.2), all such contributions are at most

$$\sum_{N \leq n} S_N B_d(1)^N \leq \exp(\tfrac{1}{2}n \log n + O_d(n)) = o(h_n n^{-A}) \quad (19.5)$$

for every fixed A . Combining the canonical characterization, (19.3) with $r = d + 2$, and (19.5) proves Theorem 19.1.

19.4 The first rooted counts

Only rooted K_2 occurs at order two, so $b_0 = 1$. At order three, P_3 has two root orbits and K_3 has one, giving $b_1 = 3$. These are the leaf, false-twin and true-twin one-vertex decorations. The six connected graphs of order four have the following root-orbit counts:

$$P_4 : 2, \quad K_{1,3} : 2, \quad C_4 : 1, \quad \text{paw} : 3, \quad \text{diamond} : 2, \quad K_4 : 1.$$

Each is a circle graph, so $b_2 = 11$. The companion independently enumerates their graph classes, representations and root orbits. These finite facts determine the first correction; their finiteness does not supply any analytic uniformity statement.

[write] Checked, 5 October 2026. A brute-force enumeration by the write over all graphs on two to five vertices gives 1, 2, 6, 21 connected classes and 1, 3, 11, 58 vertex orbits, and root-stabilizer orders $\{1:1\}$, $\{1:1, 2:2\}$, $\{1:3, 2:6, 6:2\}$ at excess 0, 1, 2, as used in Subsection 22.1. That every graph on at most five vertices is a circle graph is the companion's enumeration; the write did not repeat it. (Added after the independent check of 5 October 2026: that check formed the intersection graphs of all matchings of at most ten points and found every graph on at most five vertices among them, so the fact is now confirmed independently of the companion.)

20 Global tails and the first graph coefficient

20.1 Uniform factorial domination

A branch multiplicity vector $(a_1, \dots, a_d)$ has excess $k = \sum_j j a_j$, occupied-port count $\ell = \sum_j a_j$, and core order $N = n - k$. Its contribution to (19.2) is

$$p_N^{\mathbf{a}}(N)_\ell \prod_{j=1}^d \frac{b_j^{a_j}}{a_j!}, \quad \ell \leq N, \quad (20.1)$$

where $(N)_\ell$ is a falling factorial. Since $n \leq (d+1)N$,

$$\begin{aligned} \frac{h_N}{h_n}(N)_\ell &= \frac{n}{N} \frac{m_N}{m_n}(N)_\ell \\ &\leq (d+1)(2N)^{-k} N^\ell \\ &\leq (d+1) \prod_{j=1}^d \left[2^{-j} \left(\frac{d+1}{n} \right)^{j-1} \right]^{a_j}. \end{aligned} \quad (20.2)$$

Using $p_N^{\mathbf{a}} = O(h_N)$, the normalized term is bounded by a constant times $\prod_j \theta_j(n)^{a_j}/a_j!$, where

$$\theta_j(n) = b_j 2^{-j} ((d+1)/n)^{j-1}, \quad \theta_1 = 3/2. \quad (20.3)$$

Vectors with $\sum_{j \geq 2} (j-1)a_j \geq q+1$ have total weight $O(n^{-q-1})$: factor out that inverse power and sum the fixed exponential weights. The factorial tail $a_1 > \lceil \log n \rceil$ is smaller than every fixed inverse power. Both bounds apply across the full range of core orders down to $n/(d+1)$. A fixed-deficit expansion is used only after this tail control.

20.2 First coefficient for connected graphs

Suppose temporarily

$$p_n^a = Ph_n(1 + \alpha/n + O(n^{-2})), \quad P > 0. \quad (20.4)$$

Take $d = 2$. By (20.3), terms with two or more excess-two branches are $O(h_n/n^2)$. Write $a = a_1, b = a_2$, and restrict $a \leq \lceil \log n \rceil$ at a superpolynomial error. For $b = 0$, exact cancellation gives

$$\begin{aligned} \frac{3^a(n-a)_a h_{n-a}}{a! h_n} &= \frac{(3/2)^a}{a!} \frac{n}{n-a} \prod_{j=0}^{a-1} \frac{1 - (a+j)/n}{1 - (j+1/2)/n} \\ &= \frac{(3/2)^a}{a!} \left(1 + \frac{-a^2 + 3a/2}{n} + O\left(\frac{(a+1)^4}{n^2}\right) \right). \end{aligned} \quad (20.5)$$

Taylor's formula for $\log(1-x)$ is uniform on this logarithmic range; sums of squared offsets are $O((a+1)^3)$, and the squared linear coefficient is $O((a+1)^4)$. The prime multiplier is $P(1 + \alpha/n + O((a+1)/n^2))$. For $A \sim \text{Poisson}(3/2)$, $\mathbb{E}[-A^2 + 3A/2] = -3/2$. The sum of the $b = 0$ terms is thus

$$Pe^{3/2} h_n \left(1 + \frac{\alpha - 3/2}{n} + O(n^{-2}) \right).$$

For $b = 1$, $N = n - a - 2$, $\ell = a + 1$, and the analogous ratio to leading order is

$$\frac{p_N^a 3^a 11(N)_{a+1}}{a! h_n} = P \frac{(3/2)^a}{a!} \frac{11}{4n} \left(1 + O((a+1)^2/n) \right).$$

Summing proves

$$c_n = Pe^{3/2} h_n \left(1 + \frac{\alpha + 5/4}{n} + O(n^{-2}) \right). \quad (20.6)$$

With $P = e^{-3}$ and $\alpha = -5$ from (14.1), this is (14.2). The elementary Stirling expansion $m_n = \sqrt{2}(2n/e)^n(1 - 1/(24n) + O(n^{-2}))$ gives its alternate normalization in Theorem 14.1.

21 Components and every fixed order

21.1 The component transfer with a proved tail

For every fixed $q \geq 0$,

$$g_n = \sum_{k=0}^q g_k c_{n-k} + O(h_n n^{-q-1}). \quad (21.1)$$

For $n > 2q$, the displayed terms count exactly graphs with a component of order at least $n - q$, uniquely. Any other graph can have its whole connected components grouped into two sets of orders $s, n - s$ with $q + 1 \leq s \leq n/2$, for sufficiently large n . Take a component already in that interval,

the complement of a larger component, or greedily combine components all of order at most q . For $q = 0$ a disconnected graph directly gives such a grouping. Overcounting yields the upper bound

$$\sum_{s=q+1}^{\lfloor n/2 \rfloor} g_s g_{n-s} \leq \frac{2C^2}{n} \sum_{s=q+1}^{\lfloor n/2 \rfloor} m_s m_{n-s} = O(m_{n-q-1}/n) = O(h_n n^{-q-1}).$$

The last step is the same endpoint-dominant ratio argument as in (19.3). No automorphism factor is needed for a unique large component and its disjoint remaining graph.

At $q = 1$, $g_1 = 1$ gives $g_n = c_n + c_{n-1} + O(h_n/n^2)$, and $h_{n-1}/h_n = 1/(2n) + O(n^{-2})$. Thus

$$g_n = Pe^{3/2} h_n \left(1 + \frac{\alpha + 7/4}{n} + O(n^{-2}) \right), \quad (21.2)$$

proving (14.3).

21.2 A second connectivity term without a second prime term

Write $A = Pe^{3/2}$, $\gamma = \alpha + 5/4$ and $\eta = \gamma + 1/2$. At $q = 2$, since $g_2 = 2$,

$$g_n - c_n = c_{n-1} + 2c_{n-2} + O(h_n/n^3).$$

The factorial ratios are

$$\frac{h_{n-1}}{h_n} = \frac{1}{2n} + \frac{3}{4n^2} + O(n^{-3}), \quad \frac{h_{n-2}}{h_n} = \frac{1}{4n^2} + O(n^{-3}).$$

Using only (20.6),

$$\frac{g_n - c_n}{Ah_n} = \frac{1}{2n} + \frac{5/4 + \gamma/2}{n^2} + O(n^{-3}).$$

Divide by $g_n/(Ah_n) = 1 + \eta/n + O(n^{-2})$ to obtain

$$\frac{g_n - c_n}{g_n} = \frac{1}{2n} + \frac{5/4 + \gamma/2 - \eta/2}{n^2} + O(n^{-3}) = \frac{1}{2n} + \frac{1}{n^2} + O(n^{-3}).$$

The extra $1/n$ in the smaller-component ratio makes first-order individual expansions sufficient for the $O(n^{-3})$ remainder. No existence of an individual second coefficient is assumed.

21.3 The conditional transfer theorem

Suppose for some fixed q that

$$\frac{p_n^a}{h_n} = P \sum_{t=0}^q \alpha_t n^{-t} + O(n^{-q-1}), \quad \alpha_0 = 1. \quad (21.3)$$

Use Theorem 19.1 with $d = q + 1$. The graph-theoretic error is $O(h_n n^{-q-1})$. Only vectors with $\sum_{j \geq 2} (j-1)a_j \leq q$ matter; there are finitely many such choices of the coordinates $j \geq 2$. For each, restrict $a_1 \leq \lfloor \log n \rfloor$ by the global bound. The exact factorial ratio in (20.1) has a Taylor expansion whose coefficients are polynomials in a_1 and whose remainder is n^{-q-1} times a fixed polynomial, multiplied by $(3/2)^{a_1}/a_1!$. The same is true after multiplying by (21.3); replacing $n - k$ by n is uniform on this range. All polynomial Poisson moments are finite, so termwise summation is justified. This proves the connected transfer through order q , and (21.1) proves the all-graph transfer. Only $b_0, \dots, b_{q+1}$ and $g_0, \dots, g_q$ enter. The statement is for each fixed order, with no convergence claim and no error uniform as q grows.

21.4 Conditional second coefficients

For clarity, the next transfer calculation can be made explicit. Assume the additional prime hypothesis

$$p_n^a/h_n = P(1 + \alpha/n + \beta/n^2 + O(n^{-3})).$$

All 21 connected graph classes on five vertices are circle graphs; their root-orbit count is $b_3 = 58$, checked by complete finite enumeration in the companion. For excess k and port count ℓ ,

$$\begin{aligned} \frac{h_{n-k}}{h_n}(n-k)_\ell &= 2^{-k}n^{\ell-k} \left(1 + \frac{L_1}{n} + \frac{L_2 + L_1^2/2}{n^2} + O(n^{-3} \text{poly}(k+\ell)) \right), \\ L_1 &= k + k^2/2 - \ell k - \ell(\ell-1)/2, \\ L_2 &= k^2/2 + k(4k^2-1)/24 \\ &\quad - (\ell k^2 + k\ell(\ell-1) + \ell(\ell-1)(2\ell-1)/6)/2. \end{aligned}$$

Here the remainder has the logarithmic-range meaning just proved, not a claim uniform in unrestricted k, ℓ . For only a excess-one branches, the first two polynomials are

$$A_1 = -a^2 + 3a/2, \quad A_2 = a^4/2 - 5a^3/2 + 19a^2/8 - a/8.$$

For one excess-two branch they require only the first polynomial $B_1 = -a^2 - a/2 + 2$. For $A \sim \text{Poisson}(3/2)$,

$$\mathbb{E}A_1 = -3/2, \quad \mathbb{E}A_2 = 15/16, \quad \mathbb{E}(A_1 + A) = 0, \quad \mathbb{E}B_1 = -5/2.$$

The other retained possibilities are two excess-two branches, with leading weight $121/(32n^2)$, and one excess-three branch, with weight $58/(8n^2)$. Including the shifted prime multiplier $1 + \alpha/n + (\alpha k + \beta)/n^2$ gives

$$\frac{c_n}{Pe^{3/2}h_n} = 1 + \frac{\alpha + 5/4}{n} + \frac{\beta + 11\alpha/4 + 163/32}{n^2} + O(n^{-3}), \quad (21.4)$$

$$\frac{g_n}{Pe^{3/2}h_n} = 1 + \frac{\alpha + 7/4}{n} + \frac{\beta + 13\alpha/4 + 223/32}{n^2} + O(n^{-3}). \quad (21.5)$$

The second line adds $\gamma/2 + 5/4$ by the component identity. With $\alpha = -5$ these yield the coefficients stated in Theorem 14.2, still conditional on β .

[write] Checked numerically, 5 October 2026. The write evaluated the model sum $\sum_N Ph_N(1 + \alpha/N + \beta/N^2)[z^{n-N}]B_3(z)^N$, with $b_0, \dots, b_3 = 1, 3, 11, 58$ and $n - N \leq 120$, at $n = 1000, 2000, 4000, 8000$ in 40-digit arithmetic. Richardson extrapolation of n^2 times its second-order residual reproduces $\beta + \frac{11\alpha}{4} + \frac{163}{32}$ to within 10^{-5} for $(\alpha, \beta) = (0, 0), (-5, 0), (1, 0), (0, 1)$; in particular $-\frac{277}{32} = -8.65625$ at $\alpha = -5, \beta = 0$. By hand, $g_n = c_n + c_{n-1} + 2c_{n-2} + O(h_n/n^3)$ with $h_{n-1}/h_n = \frac{1}{2n} + \frac{3}{4n^2} + O(n^{-3})$ and $h_{n-2}/h_n = \frac{1}{4n^2} + O(n^{-3})$ turns (21.4) into (21.5). This checks the algebra of the two displays from the formula of Theorem 19.1; it does not check the transfer theorem, and it says nothing about β .

22 Bounded automorphism weights

The same canonical proof yields additional first corrections for bounded observables. For $t \in [0, \infty)$, define

$$c_n(t) = \sum_{\substack{G \text{ connected} \\ |G|=n}} |\text{Aut}(G)|^{-t}, \quad g_n(t) = \sum_{|G|=n} |\text{Aut}(G)|^{-t}.$$

Both sums range over unlabeled circle-graph isomorphism classes. At $t = \infty$, interpret the weight as the indicator of a trivial automorphism group. Every weight lies in $[0, 1]$, so excluded large-split and symmetric-core classes still have mass $O(h_n/n^2)$, uniformly in t . On the recovered asymmetric-core class, the automorphism group is the direct product of the rooted branch groups. Replace each b_j by

$$b_j(t) = \sum_{\substack{(S,\rho) \text{ rooted connected circle graph} \\ |S|=j+2}} |\text{Aut}(S, \rho)|^{-t}.$$

Then $0 \leq b_j(t) \leq b_j(0)$ and all the factorial bounds remain uniform. This is a bounded-weight argument; it does not control positive powers of the automorphism-group order.

22.1 Weighted transfer and small branch groups

With general b_1, b_2 the Poisson parameter is $b_1/2$. The expectation of $-A^2 + 3A/2$ is $-b_1^2/4 + b_1/4$, and one excess-two branch contributes $b_2/4$. Consequently, uniformly in $t \in [0, \infty]$,

$$c_n(t) = e^{-3+b_1(t)/2} h_n \left(1 + \frac{-5 + (b_2(t) + b_1(t) - b_1(t)^2)/4}{n} + O(n^{-2}) \right). \quad (22.1)$$

The uniformity follows from the already uniform graph error, the prime error independent of t , and bounded Poisson moments for $0 \leq b_1(t) \leq 3$. In an all-graph decomposition the unique large component cannot be exchanged with its smaller complement, so the automorphism weight multiplies. The singleton complement has weight one, giving $g_n(t) = c_n(t) + c_{n-1}(t) + O(h_n/n^2)$; its relative coefficient in (22.1) increases by $1/2$.

The three excess-one rooted branch groups have orders 1, 2, 2. The eleven excess-two groups have orders 1 in three cases, 2 in six cases, and 6 in two cases. The rigid cases are the two root orbits of P_4 and the paw rooted at an interchangeable degree-two vertex. The order-six cases are the star rooted at its center and the clique K_4 , both with group S_3 ; all order-two groups are C_2 . Thus

$$b_1(t) = 1 + 2 \cdot 2^{-t}, \quad b_2(t) = 3 + 6 \cdot 2^{-t} + 2 \cdot 6^{-t}. \quad (22.2)$$

These small group counts are exhaustively checked in the companion.

22.2 Labeled graphs and asymmetry

At $t = 1$, $b_1 = 2$, $b_2 = 19/3$, and $(b_2 + b_1 - b_1^2)/4 = 13/12$. Multiplying the reciprocal- automorphism sum by $n!$ counts labeled graphs on a prescribed n -element vertex set. Therefore

$$C_n^{\text{lab}} = n! e^{-2} h_n \left(1 - \frac{47}{12n} + O(n^{-2}) \right), \quad (22.3)$$

$$G_n^{\text{lab}} = n! e^{-2} h_n \left(1 - \frac{41}{12n} + O(n^{-2}) \right). \quad (22.4)$$

These are mathematical consequences of the bounded-weight proof; no separate labeled OEIS identification or exhaustive labeled literature-priority audit is claimed.

At $t = \infty$, $b_1 = 1$, $b_2 = 3$, so

$$c_n(\infty) = e^{-5/2} h_n \left(1 - \frac{17}{4n} + O(n^{-2}) \right), \quad g_n(\infty) = e^{-5/2} h_n \left(1 - \frac{15}{4n} + O(n^{-2}) \right).$$

Under either uniform connected or uniform arbitrary unlabeled sampling, division by the respective ordinary count gives

$$\mathbb{P}(\text{Aut}(G) = 1) = e^{-1} \left(1 - \frac{1}{2n} + O(n^{-2}) \right). \quad (22.5)$$

More generally put $\rho = 2^{-t}$ and $\sigma = 6^{-t}$, with both zero at infinity. Equations (22.1)–(22.2) give, uniformly in $t \in [0, \infty]$ under either sampling law,

$$\mathbb{E}[|\text{Aut}(G)|^{-t}] = e^{\rho-1} \left(1 + \frac{-1/2 + \rho - \rho^2 + \sigma/2}{n} + O(n^{-2}) \right). \quad (22.6)$$

At $t = 1$ this is $e^{-1/2}(1 - 1/(6n) + O(n^{-2}))$. The universal prime coefficient cancels from the normalized transform, although its controlled first-order expansion underlies the argument.

22.3 Rare nonabelian groups

Through first order, all recovered branch groups are trivial or C_2 , apart from the two excess-two branches with group S_3 . Two excess-two branches, larger branch types and excluded splits have probability $O(n^{-2})$. Replacing the single excess-two count 11 by 2 in Section 20 gives

$$\mathbb{P}(\text{Aut}(G) \text{ nonabelian}) = \frac{1}{2n} + O(n^{-2}). \quad (22.7)$$

The same estimate holds for not being an elementary abelian 2-group. For each fixed $k \geq 0$, the refined law is

$$\mathbb{P}(\text{Aut}(G) \cong S_3 \times (C_2)^k) = \frac{e^{-1}}{2n k!} + O(n^{-2}) \quad (22.8)$$

under either connected or arbitrary unlabeled sampling. Indeed, the two C_2 excess-one branch types have total limiting Poisson mean one, the rigid type has mean $1/2$, and one S_3 branch contributes $2/(4n)$. Summing rigid branches and selecting exactly k nonrigid excess-one branches gives the factor $e^{-1}/k!$ after normalization. For arbitrary sampling, the only disconnected class at order $1/n$ is a giant connected graph plus one isolated vertex. Its nonabelian subfamily pays a further $1/n$ factor and so has probability $O(n^{-2})$; all other disconnected exceptions are already $O(n^{-2})$. This justifies the same rare-group coefficients for arbitrary sampling. This fixed- k statement is not an unbounded automorphism moment claim.

23 Refined least index inverses

For $a_n = c_n$ or g_n , define

$$N_a(y) = \min\{n \geq 1 : a_n \geq y\}.$$

Both sequences are unbounded and eventually strictly increasing. Indeed Theorem 14.1 gives $a_{n+1}/a_n \sim h_{n+1}/h_n \sim 2n$, without differentiating any unknown error. Their finite pre-onset prefixes are bounded, so for sufficiently large y the least threshold lies in the increasing tail. This is enough for the asymptotic inverse; no separate finite-order monotonicity assertion is required.

Let $L = \log y$, and use the positive real Lambert branch W_0 :

$$u = \frac{L}{W_0(2L/e)}, \quad D = \log(2u), \quad B = \frac{3 + \log 2}{2}. \quad (23.1)$$

Then $u(\log(2u) - 1) = L$. For the connected and all-graph cases respectively set

$$K_c = \frac{103}{24}, \quad K_g = \frac{91}{24}, \quad s_a(y) = u + 1 + \frac{B}{D} + \frac{K_a - B^2/(2D^2)}{uD}.$$

Theorem 23.1 (Ceiling safe first refinement). *For each of $a = c, g$, there exist constants $C_a, y_a > 0$ such that, for every $y \geq y_a$,*

$$\left\lceil s_a(y) - \frac{C_a}{u^2 D} \right\rceil \leq N_a(y) \leq \left\lceil s_a(y) + \frac{C_a}{u^2 D} \right\rceil. \quad (23.2)$$

The centers satisfy $s_c(y) - s_g(y) = 1/(2uD)$. The bracket does not authorize unconditional rounding at inputs where its interval meets an integer. Its constants and onset are noneffective in this proof.

Proof. Write $a_n = e^{-3/2} h_n(1 + \delta/n + O(n^{-2}))$, with $\delta = -15/4$ or $-13/4$. The smooth positive normalization is

$$H(t) = \frac{e^{-3/2} 2^t \Gamma(t + 1/2)}{4\sqrt{\pi} t}, \quad F_\delta(t) = \log H(t) + \frac{\delta}{t}.$$

Stirling's expansion gives

$$\log H(t) = t(\log(2t) - 1) - \log t - \frac{3(1 + \log 2)}{2} - \frac{1}{24t} + O(t^{-3}). \quad (23.3)$$

Its derivative is obtained from the smooth gamma function, so $F'_\delta(t) = \log(2t) - 1/t + O(t^{-2}) > 0$ eventually. Also $\log a_n = F_\delta(n) + O(n^{-2})$; no derivative is taken of this integer-sequence remainder.

For bounded d , expanding at u gives

$$F_\delta(u + d) - L = Dd - \log u - \frac{3(1 + \log 2)}{2} + \frac{d^2/2 - d - 1/24 + \delta}{u} + O(u^{-2}).$$

Since $\log u = D - \log 2$, the order-one solution is $d_0 = 1 + B/D$. Put $d = d_0 + d_1/u$. The next coefficient vanishes when

$$d_1 = \frac{13/24 - \delta - B^2/(2D^2)}{D}.$$

Thus $K_a = 13/24 - \delta$, giving the claimed values. Substitution has residual $O(u^{-2})$. The unique large root of $F_\delta(t) = L$ differs from $s_a(y)$ by $O(1/(u^2 D))$, because the derivative is comparable with D near u .

To pass to the integer sequence, first a coarse leading bracket locates the threshold and the smooth roots in $[u/2, 2u]$ for large y . The tail supremum of $|\log a_n - F_\delta(n)|$ over integers $n \geq \lfloor u/2 \rfloor$ is $O(u^{-2})$. Let r_- and r_+ solve $F_\delta(r_\pm) = L \pm C'u^{-2}$, with C' large enough. If $n < \lceil r_- \rceil$ in this tail then $F_\delta(n) < L - C'u^{-2}$, so $a_n < y$; at $n = \lceil r_+ \rceil$, $a_n \geq y$. The bounded prefix is already below y . Hence $\lceil r_- \rceil \leq N_a(y) \leq \lceil r_+ \rceil$. Both roots differ from $s_a(y)$ by $O(1/(u^2 D))$, proving (23.2) after increasing its constant. $\square$

Higher prime coefficients, if proved with controlled errors, could be passed through the fixed-order transfer and then inverted recursively by the same smooth procedure. The displayed theorem uses only the unconditional first corrections. It makes no unconditional all-orders inverse claim.

[write] The inverse, 5 October 2026. At $x = y$, $s_a(y) = s(x) + (K_a - B^2/(2D^2))/(uD)$, with s Part I's centre (10.12). The leading u is an instance of the transseries volume's factorial core, and the bracket follows the pattern of its staircase theorem without being an instance (front matter).

24 Reproducibility and limits

The companion is a fresh exact implementation for this report. Its role is to verify finite atom, graph and rational identities, not to certify analytic asymptotics from samples. In particular it checks local overlap and compatible-cluster counts, the four three-chord interval cases, rooted graph-isomorphism counts, first and conditional second transfer arithmetic, the inverse coefficient cancellation, and small rooted automorphism orders. The README records the exact tested ranges and outputs. Explicit exceptions, rather than removable Python assertions, enforce the checks. Mutation tests require incorrect coefficients to fail, including under Python’s optimization mode; output-safety tests reject overwrites and symlink routes.

The package supplies editable LaTeX, the rendered PDF, exact companion source and evidence, source provenance, checksums, deterministic builders and release tests. It excludes downloaded papers, exploratory working notes, and earlier report files. PDF byte reproducibility is qualified to the tested pdfTeX installation; the mathematics does not depend on identical PDF bytes. ZIP byte reproducibility uses fixed entry order, timestamps, permissions and stored payloads. A checksum manifest checks consistency, not authenticity against joint replacement of a file and its hash.

The established new content is the uniform local first correction, its direct $O(n^{-2})$ nonlocal comparison, the canonical finite split-tree transfer, the individual graph first coefficients, the second connectivity term, and the refined ceiling-safe inverse. An unknown prime second coefficient remains unknown. Effective constants, a final-journal source reconciliation, and any stronger priority claim remain outside the proved scope. The bounded reciprocal-automorphism refinements and labeled consequences in Section 22 are included with their proof; unbounded automorphism moments are not claimed.

[write] **Status, 5 October 2026.** The unknown β and the other limits stated here are collected, with Part I’s, in Section 25. The companion is shipped as `code/152-corr-companion-*.py` and `data/152-corr-companion-*`; at intake its checks were rerun on a copy and the recorded result regenerated byte for byte (README). The `SOURCE_PROVENANCE.json` entries on “independent review” are metadata, as that file says, not an independent proof certificate.

25 Further questions and research

[write] **Standing rule, 5 October 2026.** This section is added in the write. Under Vladimir’s standing rule of 4 October 2026 it collects every question left open by the two manuscripts, with its source, the argument sketch it came with, and what is missing. Part I’s Section 12 item 1 is answered by Part II (dated note there). Nothing in the two manuscripts was found to be wrong, and no claim was refuted.

1. *The prime second coefficient β* (Part II, Theorem 14.2, Subsection 21.4, Section 23). If $p_n^a = e^{-3}h_n(1 - \frac{5}{n} + \frac{\beta}{n^2} + O(n^{-3}))$, then the second relative coefficients of c_n and g_n are $\beta - \frac{277}{32}$ and $\beta - \frac{297}{32}$ (proved conditionally), and the n^{-2} term of c_n/g_n does not depend on β (proved). Part II gives no value and no conjecture. What is missing is the next order in each step of Sections 16–18: the polymers of four chords, whose total activity Subsection 16.2 bounds by $O(M^{-2})$, the hard-core expansion (17.1) beyond its pair term, the matching probabilities (18.1) beyond r^2/M , and the nonlocal term: zero-defect matchings with a k -decomposition, $4 \leq k \leq n - 4$, have probability $O(n^{-2})$ by the bound before (18.3), so at second order they must be counted rather than bounded. The intake Monte Carlo concerns only the first-order constant and cannot resolve a second-order term at its sample sizes.

2. *An unconditional expansion to every fixed order* (Part I, Section 12 item 2; Part II, Theorem 14.2 and Subsection 21.3). Part II reduces every fixed order of c_n and g_n to the same order of p_n^a/h_n . It sketches that higher prime coefficients “could be passed through the fixed-order transfer and then inverted recursively by the same smooth procedure” (Section 23); what is missing is the prime expansion itself at each order (item 1 is its first case). Neither Part claims convergence, a remainder uniform in the order, or an all-orders inverse.
3. *Effective constants, onsets and certified thresholds* (Part I, item 3 and the remarks after Theorem 9.1; Part II, scope paragraph and Theorem 23.1). Only the local bound $\min\{1, 200/n\}$ of (9.11) is explicit. The graph-count constants, the constant of (3.6) and (15.3), the onsets, and the inverse constants C_* , C_a exist but are not computed, so neither bracket decides the least index at a given input. What is missing is an explicit constant in every O -term of one complete chain; Part II’s chain does not use the qualitative limit (2.3), so it is the natural candidate.
4. *Source reconciliation and priority* (Part I, item 4 and Subsection 2.3; Part II, Subsection 15.2). The 2026 journal text of [1] was not read at intake or at the write; Gabor, Supowit and Hsu’s paper was read only through [1, Proposition 5.12], and the paper of Arratia, Bollobás, Coppersmith and Sorkin at abstract level. A stronger priority statement than “not located in the sources checked” needs these.
5. *Labeled counts* (Part I, item 4 and the paragraph after (8.8); Part II, after (22.4)). The labeled equivalents and first corrections are proved; a literature and OEIS identification audit for labeled circle graphs was made by neither manuscript, nor by the write.
6. *Unbounded automorphism observables* (Part I, Proposition 8.2 and Section 9; Part II, Section 22). Both Parts control only bounded functions of $|\text{Aut}(G)|$ and say so. The limiting law 2^P , $P \sim \text{Poisson}(1)$, has $\mathbb{E} 2^{tP} = \exp(2^t - 1) < \infty$ for every t ; whether $\mathbb{E} |\text{Aut}(G)|^t$ converges to it is open. What is missing is uniform integrability: control of the rare graphs with large automorphism groups (symmetric cores, large splits), which the bounded-weight arguments discard without bounding their weight.
7. *Exact counts beyond order 13* (write). The exact data stop at g_{13} (Johnston, 2020) and the derived c_{13} . At $n = 13$ the counts are still 10.5% (c_{13}) and 11.3% (g_{13}) below the first-corrected approximations (dated note in Section 11), so the expansions cannot yet be tested numerically, nor β estimated. Part II’s Theorem 19.1 expresses c_n through asymmetric prime counts and rooted branch counts up to an error; an exact version, counting all prime centres with their port actions, would give a recurrence from the counts of prime circle graphs. Neither manuscript states one.

[write] **Independent check of the write, 5 October 2026.** An adversarial check made by the intake after the write (commit c6557f9c9) reread printed p. 23 of arXiv:2402.06394v1 (the file with the recorded SHA-256) and confirmed Remark 3.3 by brute force over all perfect matchings of $2n$ points, $n \leq 7$, against each of the $4n - 1$ nonidentity dihedral symmetries: a half-turn and every reflection through gaps fix $t_n = 3, 7, 25, 81, 331, 1303$ matchings for $n = 2, \dots, 7$, a reflection through two endpoints fixes t_{n-1} , and $S_n = 1, 3, 15, 89, 465, 2307, 11823$ for $n = 1, \dots, 7$, below (3.3). It found that the source’s relabelling is right only when read with $2n + 1 - i$ (sentence added to the remark). It also confirmed the sharpening $e^{-4} \rightarrow e^{-3}$ from the source’s Lemmas 5.7–5.8 and Proposition 5.11, and the quoted bound (12). It confirmed the live OEIS entries (A156808 #9; A156809 #16, with Johnston’s $a(13)$) and Johnston’s post, and $c_{13} = 21,593,488,017$ by Euler inversion, whose Euler transform returns A156809 through $n = 13$. It confirmed every ratio of the

notes in Section 11, the percentages of item 7, and the binomial standard errors of the Monte Carlo note at Theorem 16.1 (the simulation was not rerun). By exhaustive enumeration it confirmed the rooted counts $b_0, \dots, b_3 = 1, 3, 11, 58$ and the root-stabilizer orders; the same run showed that every graph on at most five vertices is a circle graph (note in Subsection 19.4). Finally it confirmed the sum 353.9617 at $M = 12$ behind (9.11), the factorial-core and Lambert-core identifications of the front matter, and that the brackets are not instances of `p0:thm:staircase` ($H(n) = e^{-3/2}h_n \neq a_n$). One wording was corrected, with a dated note keeping the first wording: the side condition $\log(2u) > 1$ of the factorial core holds for every $L > 0$, not only for large L . No mathematical error was found.

References

- [1] F. Bassino, M. Bouvel, V. Féray, L. Gerin and A. Pierrot, *Dense and nondense limits for uniform random intersection graphs*, arXiv:2402.06394v1, 9 February 2024. [Versioned arXiv record](#). Part I: Sections 5.2–5.3 and Appendix A.2 are the primary inputs. The 2026 journal metadata is identified in the text; its final text was not inspected. Part II: Sections 5.2–5.3 and Appendix A.2 are the inspected primary circle-graph inputs; the final 2026 journal text was not inspected. **[write]** Journal version: *Electronic Journal of Probability* **31** (2026), article 110, [doi:10.1214/26-EJP1570](https://doi.org/10.1214/26-EJP1570); not read.
- [2] L. E. Danielsen and M. G. Parker, *Interlace Polynomials: Enumeration, Unimodality, and Connections to Codes*, arXiv:0804.2576v2, 1 October 2009, Section 3, Table 3. [Versioned arXiv record](#).
- [3] E. Gioan and C. Paul, *Split decomposition and graph-labelled trees: Characterizations and fully dynamic algorithms for totally decomposable graphs*, Discrete Applied Mathematics **160** (2012), 708–733. [doi:10.1016/j.dam.2011.05.007](https://doi.org/10.1016/j.dam.2011.05.007). Section 2.2, especially Theorem 2.9 and Corollary 2.10. **[write]** Also arXiv:0810.1823v2.
- [4] P. Klavík and P. Zeman, *Automorphism Groups of Geometrically Represented Graphs*, 32nd International Symposium on Theoretical Aspects of Computer Science (STACS 2015), LIPIcs **30**, 540–553, 2015. [doi:10.4230/LIPIcs.STACS.2015.540](https://doi.org/10.4230/LIPIcs.STACS.2015.540). Part II: Section 3, circle graphs, canonical split trees and prime automorphisms. **[write]** Also arXiv:1407.2136.
- [5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A156808, number of nonisomorphic connected circle graphs of order n ; inspected 3 October 2026 (Part I). <https://oeis.org/A156808>.
- [6] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A156809, number of nonisomorphic circle graphs of order n ; inspected 3 October 2026 (Part I). <https://oeis.org/A156809>.
- [7] **[write]** *Transseries and Inversion*, the transseries volume of this repository, Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex; p0:prop:factorial-core, p0:thm:lambert-core, p0:thm:staircase.